Complex Hamiltonian learning and KAM-preserving magnetic control of a catheter model

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Abstract

A learned magnetic-catheter model should connect prediction errors to bounded motion and admissible actuation. We study this connection for a clamped–free beam and a specified planar Kirchhoff restriction. For a finite-feature complex Hamiltonian network, nonzero-loss COMSOL identification yields coefficient and analytic-error bounds supporting an a posteriori KAM proof at approximately 1.9 mm amplitude. Magnetic force-and-torque feedback realizes this conservative motion in the damped 64-mode plant, with a whole-centreline bound below 1.971 mm for zero higher-mode startup. An input-work constraint extends clearance to driven motion under stated sensing and actuation errors. An S-shaped endpoint task achieves $6.419\,\mu$m RMS tracking error. Two complex fully connected hidden layers with pole-constrained hyperbolic tangents provide an end-to-end extension. Its frozen fixed-mass conservative energy has a separately certified invariant torus and local function-error bounds; a nominal torus at the same frequency has a quantified phase-paired distance and a measurable amplitude discrepancy. For configuration-dependent inertia, we derive the mechanical momentum, kinetic normal form and matched dissipative Birkhoff structure. A planar COMSOL experiment identifies a complex potential and scalar drag, then trains a separate complex PSD factor network with that energy frozen. Weak-loss descent, whole-box field-error bounds and an exact port-power identity apply to this decoder. Validated reconstructions connect its native observations to continuous weak equations for the same finite-Gauss target. A true-power filter proves continuous-time safety for that chart and is checked numerically. For the same frozen planar fit, continuous-data error bounds and an explicit parameter response certify same-frequency invariant tori at every real point of the conservative learned–reference homotopy. Statistical generalization remains unverified. The results distinguish conservative persistence, dissipative power balance and control safety; the comparisons do not establish a general advantage of complex over real networks.

Keywords

magnetic catheter robots, complex Hamiltonian learning, symplectic integration, KAM theory, magnetic control

1 Background and research question

Magnetic actuation can steer a flexible catheter without placing motors along its body. A field applies torque to a magnetic segment, and a field gradient can supply force; elastic deformation and dissipation then determine the motion of the entire device (Abbott et al. 2020; Kim et al. 2019). For operation inside a confined passage, a useful dynamical model must therefore do more than predict the tip. It must connect available magnetic inputs to bounded centreline motion. This paper develops that connection for a clamped–free catheter model, using a learned Hamiltonian as the common object for numerical integration, nonlinear stability analysis and feedback design.

The application motivation is remote manipulation in reduced gravity. Spaceflight creates medical and operational conditions that warrant dedicated investigation; NASA's assessment of spaceflight-associated neuro-ocular syndrome is one example of a gravity-dependent health risk (NASA 2025). For a remotely manipulated device in this setting, we isolate the following mechanical question: how to sustain a prescribed class of bounded motions and respect a clearance limit when a flexible magnetic device has distributed inertia, elasticity and damping. The model is planar, has small slopes and neglects gravity and contact. These assumptions make the energy balance explicit and permit independent finite-element calculations, while identifying the physical effects that a later device study must supply.

Two operating objectives organize the analysis. The first is recurrent motion with a proved amplitude bound, represented by an invariant torus of a conservative model. The second is tracking a requested endpoint path while keeping the intervening body inside a corridor. The first objective asks whether motion persists for arbitrarily long times; the second asks how much external work can safely be supplied during a manoeuvre. A common mechanical energy links them, but they call for different conclusions. We use KAM theory for the recurrent motion and an input-work inequality for clearance, then derive magnetic commands that make those conclusions relevant to the same damped beam.

The main difficulty is preserving this connection through learning and actuation. Low prediction error alone leaves the Hamiltonian's nonlinear derivatives undetermined. Even with accurate derivatives, a numerical solver can introduce an error comparable to the difference between two learned models. Finally, damping removes the recurrent motion of a conservative system unless a controller supplies the appropriate work. We address these issues in sequence: construct a power-consistent learned energy, resolve its approximation and integration errors, establish an invariant motion of the fitted function, and realize that motion with the available magnetic force and torque.

Recent magnetic-robot control studies address shape regulation and lumen navigation directly. Francescon et al. (2025) combine a pseudo-rigid-link model with vision feedback and homogeneous fields to control a robot with a prescribed lengthwise magnetization profile, demonstrating navigation in a soft pancreatic-duct phantom under varied insertion poses. Tong et al. (2025) use a geometrically nonlinear magnetic-rod simulation with reduced contact handling to compute fields that reduce distal-tip constraint forces, with planar phantom experiments and three-dimensional numerical demonstrations. These studies establish practical routes from a mechanical model to magnetic commands. The question pursued here is how a learned energy, with a quantified identification error, can support both persistent motion and bounds on the entire centreline. For the stated planar beam, we connect analytic error and twist bounds to a finite-amplitude invariant torus, realize its low-mode dynamics in the damped full finite-element plant, and bound driven motion through supplied work. Table 1 summarizes the objects passed between these steps.

1.1 Learning a mechanical model that can be analyzed

The starting point is a mechanical energy with explicit boundary work. Variational beam and rod formulations relate distributed strain energy, inertia and applied loads before discretization (Simo 1985; Antman 2005). In the present specialization, the tip magnet contributes a cosine potential under a static bias field. Its curvature enters the linear modal frequencies, while its higher-order terms govern the amplitude dependence of those frequencies. We retain the complete cosine because the nonlinear part later supplies the twist required for KAM nondegeneracy. Modal coordinates are normalized once, and the same dimensional map is used to interpret learned coefficients, simulated trajectories and magnetic commands. Supplement S1 supplies the distributed derivation; Section 2 develops the model used in the analysis.

Hamiltonian neural networks provide a natural learning mechanism: fit a scalar energy and obtain its canonical vector field by differentiation (Greydanus et al. 2019). This makes the energy derivatives available to both numerical integration and stability analysis. A magnetic catheter also loses and receives energy, so we follow the separation of conservative dynamics, dissipation and inputs used by dissipative SymODEN and port-Hamiltonian networks (Zhong et al. 2020; Desai et al. 2021). The learned conservative part specifies reversible dynamics; a nonnegative dissipation model and explicit magnetic ports complete the physical decoder. This separation is necessary for a controller to distinguish the energy stored by the device from the work it must supply.

Complex coordinates package modal position and momentum into one state. This representation supports shared complex features (Trabelsi et al. 2018). Their physical interpretation remains tied to the underlying real energy. We derive that representation explicitly, including the conjugate dependence and the analytic extension used in the KAM argument. Complex-network approximation theory offers useful existence and comparison results, but those results depend on the activation, target class and norm (Zhang et al. 2022). Here the relevant question is whether the fitted model controls the required derivatives on a declared domain. Smooth approximation theory (Hornik 1991) provides an existence route; the quantitative finite-feature analysis uses the actual feature class and its fitted coefficients. An end-to-end extension learns two complex fully connected hidden layers with constrained complex $\tanh$, and separates empirical prediction tests from the additional analytic checks needed for its own KAM certificate.

Table 1. Four steps of the study. Each step supplies the object needed by the next; the detailed proofs and additional experiments are in the Supplement.

Step	Question	Result passed to the next step
1. Derive, Section 2	Which energy and discrete update describe the beam?	A power-consistent complex representation and a complete central symplectic update.
2. Evaluate, Section 3	What errors remain after fitting and integration?	Paired comparisons and quantitative approximation bounds for a specified learned function.
3. Establish persistence, Section 4	Does that function preserve a nonlinear recurrent motion?	Nondegenerate twist, a finite-amplitude invariant torus and a local family of tori.
4. Realize, Section 5	How does the motion guide magnetic actuation?	Force/torque feedback, full-model amplitude bounds and clearance-constrained tracking.

That feature class combines the complete magnetic potential with an analytic polynomial residual. Its coefficients can be fitted from energy derivatives or state increments, and all derivatives needed for the subsequent analysis are explicit. A residual may depend on momentum as well as position. We retain that dependence because it changes both the Hamiltonian vector field and the relation between physical velocity and canonical momentum. The magnetic feedback is consequently derived from the full learned gradient and Hessian. This gives learning a concrete role: it supplies a differentiable energy whose approximation error can be bounded and whose dynamics can be translated back into physical commands.

The distinction between identification and model reduction is particularly important for this device. A low-mode projection of a full beam trajectory need not satisfy a closed low-mode equation: the discarded modes also contribute to tip rotation and hence magnetic torque. A fitted residual can absorb part of that discrepancy on the observed trajectories. Its predictive value should therefore be assessed against independent full-beam solutions, while a theorem about that residual must use the function actually obtained by fitting. We organize the computations around both questions. Controlled reduced-model data isolate coefficient recovery and permit a quantitative learning-to-KAM argument. Independent full finite-element observations assess the discrepancy introduced by reduction and the response to driven motion.

The comparison design separates the effects of representation and mechanical knowledge. Real and complex implementations are fitted with and without the known quadratic energy, using paired training seeds and the same held-out trajectories. Five paired contrasts measure the effect of the prior in each representation, the representation effect with and without that prior, and their interaction. A separate integration comparison resolves numerical error before small learning differences are interpreted. The resulting real and complex predictions are equivalent at the resolved precision for this feature class. The deep extension reports the same five contrasts separately, using matched held-out data; changing the trainable function class requires a new comparison.

1.2 *From numerical trajectories to persistent physical motion*

A learned Hamiltonian is continuous in time, whereas prediction and control require discrete updates. Central generating functions provide classical symplectic difference schemes for this step (Feng et al. 1989). We derive the fourth-order central formula through its midpoint expansion and specialize it to the fitted magnetic energy. The correction, its symplecticity and its order belong to the established geometric-integration framework (Hairer et al. 2006); the model-specific work is the complete realization for the learned, generally nonseparable Hamiltonian and its numerical evaluation. This treatment exposes the derivatives actually required by the update and allows the solver tolerance, phase error and state error to be assessed together.

The resulting numerical trajectories motivate the stability question, but KAM persistence is determined by properties of the Hamiltonian itself. Classical KAM theory connects sufficiently small perturbations of nondegenerate integrable systems to families of invariant tori (Pöschel 2001; Bounemoura 2020). For the magnetic beam, the nonlinear frequency variation can be computed from the retained cosine. Learning must preserve that variation and control the full remainder. The analysis therefore follows coefficients and analytic norms through to the theorem assumptions.

Chen et al. connect nonzero Hamiltonian-network training loss to KAM theory under statistical and regularity hypotheses (Chen et al. 2022). We develop a computable counterpart for the present analytic feature class. In the experiment used for the finite-amplitude proof, state-only COMSOL observations determine an integral regression. Bounds on its residual, integration defect and design matrix yield a coefficient-error bound; differentiating the same features converts it into analytic-error and twist bounds. Thus the learning error remains part of the argument even

though the training loss is nonzero. The broader statistical route is retained in Supplement S3, while the main proof uses the deterministic bounds appropriate to these trajectory observations.

With those bounds in place, an a posteriori KAM theorem tests a computed torus of the fitted Hamiltonian through its invariance defect and geometric conditions (Haro and Luque 2019). We apply it to the exported learned function, including its complete cosine and momentum-dependent residual. The result is a torus at approximately 1.9 mm amplitude. Nondegeneracy of its final torsion also supplies a local family of positive measure through an established torus-stability result (Bounemoura 2016). Explicit measure estimates require a specified domain and constants (Biasco and Chierchia 2018; Chierchia and Koudjinan 2021); we retain that calculation in the Supplement. Weighted Birkhoff averages additionally explore a wider domain over finite times (Das et al. 2017; Duignan and Meiss 2023), providing a complementary numerical view of the recurrent motions.

For these motions to guide actuation, the conservative subsystem must be connected to the damped full beam. We use a transverse magnetic force and torque to reproduce the learned low-mode accelerations. The remaining modes then form stable forced oscillators, whose quasiperiodic responses can be constructed and bounded. This yields an invariant torus of the complete 64-mode model and a bound that also includes startup from zero higher-mode states. Leaving the torus for an endpoint task invokes a second control mechanism: a magnetic-work governor limits physical energy and hence the whole-centreline deflection. The two mechanisms address recurrent operation and manoeuvring within the same mechanical setting.

1.3 Contributions and study design

The finite-feature learning-to-control chain contributes three model-specific results. First, it derives the complex magnetic Hamiltonian, implements the classical fourth-order central update for the learned nonseparable energy, and resolves representation, prior and integration effects through matched comparisons. Second, it converts nonzero state-only identification error into computable analytic and twist bounds, then verifies a finite-amplitude invariant torus of that fitted function. Third, it constructs magnetic feedback and a full-model torus lift, together with an energy-based clearance condition for driven motion. The general approximation and KAM theorems are established results, cited with their hypotheses and identifiers where applied; the coefficient bounds, mechanical constructions and application proofs are developed here.

The quantitative results make the connection operational within the stated model. Starting the low state on the certified torus and all higher states at zero gives a whole-centreline bound below 1.971 mm, a transverse-field bound below 0.922 mT and a gradient bound below 0.02573 T/m. A separate S-shaped endpoint task achieves 6.419 μm RMS error with the learned controller.

The deep extension adds a trainable hidden representation, joint energy–damping identification, and a conditional analytic transfer argument. Its domain and derivative diagnostics concern the frozen deep weights, whereas the verified torus and full-beam guarantees above concern their specified finite-feature fits.

Two COMSOL identification settings serve distinct purposes within this common mechanical model: full finite-element observations test prediction and tracking, whereas the two-mode integral observations supply the learning-to-KAM proof and the associated magnetic realization. Their fitted coefficients and error budgets are specified separately. Section 6 brings the results together.

2 Derivation of the complex Hamiltonian and symplectic scheme

2.1 From magnetic rod mechanics to canonical variables

We first derive an energy from magnetic beam mechanics, express that energy in complex coordinates, and then discretize its conservative flow. For a planar, small-slope cantilever, the transverse displacement $w(s,t)$ satisfies

$$\begin{aligned} \rho A_c w_{tt} + c_d w_t + EI w_{ssss} = 0, \qquad & 0 < s < L, \\ w(0,t) = w_s(0,t) = 0, \qquad & w_{sss}(L,t) = 0, \\ EI w_{ss}(L,t) = \mu[B_y \cos\vartheta - B_0 \sin\vartheta], & \end{aligned} \tag{1}$$

where $\vartheta = w_s(L,t)$, $A_c = \pi r_c^2$, and $I = \pi r_c^4/4$. The negative rotation derivative of the magnetic potential $-\mu(B_0\cos\vartheta + B_y \sin\vartheta)$ gives the tip moment. Thus the free-end rotation remains unconstrained, the constant bias B_0 stores energy, and the transverse field $B_y(t)$ supplies work.

The 80 mm beam has radius 0.6 mm and uses 32 consistent-mass cubic-Hermite elements, leaving 64 free coordinates after clamping; supplementary Table S1 gives its material and field parameters. Let q, M, K and e_ϑ denote the nodal vector, mass matrix, bending stiffness and tip-rotation selector. Two integrations by parts in the Galerkin weak form place the natural moment on e_ϑ, giving the finite-element

equations and stored energy

$$M\ddot{q} + \alpha M\dot{q} + Kq = \mu e_\vartheta [B_y \cos\vartheta - B_0 \sin\vartheta],$$
$$H_0(q,p) = \tfrac{1}{2} p^{\mathsf{T}} M^{-1} p + \tfrac{1}{2} q^{\mathsf{T}} K q + E_0[1 - \cos(e_\vartheta^{\mathsf{T}} q)], \tag{2}$$

where $p = M\dot{q}$, $E_0 = \mu B_0 = 6.0 \times 10^{-7}$ J and $c_d = \alpha\rho A_c$, with $\alpha = 0.55558524\,\mathrm{s}^{-1}$. The cosine potential remains exact within the small-slope beam model.

The bias-field stiffness is $K_b = K + E_0 e_\vartheta e_\vartheta^{\mathsf{T}}$. Choose $\Phi_n^{\mathsf{T}} M \Phi_n = I$ and $\Phi_n^{\mathsf{T}} K_b \Phi_n = \mathrm{diag}(\omega_j^2)$ for n retained modes, and write $q = \Phi_n a$, with reduced canonical momentum $\pi = \dot{a}$. Set $\omega_0 = \omega_1$, $\tau = \omega_0 t$, and $\nu_j = \omega_j/\omega_0$. Define the scalar position scales $S_j = \sqrt{E_0/(\omega_0\omega_j)}$. The balanced scaling is

$$a_j = \sqrt{\frac{E_0}{\omega_0\omega_j}} X_j, \qquad \pi_j = \sqrt{\frac{E_0\omega_j}{\omega_0}} Y_j. \tag{3}$$

The scale product E_0/ω_0, energy H_0/E_0 and time $\tau = \omega_0 t$ give canonical dimensionless equations. Define $S = \Phi_n\,\mathrm{diag}(S_j)$, $d = S^{\mathsf{T}} e_\vartheta$, and $\ell(s)^{\mathsf{T}} = N(s)S$, where $N(s)$ is the Hermite displacement row. Then $w(s) = \ell(s)^{\mathsf{T}} X$, $\vartheta = d^{\mathsf{T}} X$, and

$$h_*(X,Y) = h_2(X,Y) + g(d^{\mathsf{T}} X),$$
$$h_2 = \tfrac{1}{2}\sum_{j=1}^{n} \nu_j (X_j^2 + Y_j^2),$$
$$g(s) = 1 - \cos s - \tfrac{1}{2} s^2. \tag{4}$$

Subtracting $s^2/2$ avoids counting the bias stiffness twice. For the two-mode learning model, $\omega_0 \approx 11.417287283\,\mathrm{rad\,s}^{-1}$, $\nu \approx (1, 6.152117504)$, $d \approx (0.225670773, 0.321798806)^{\mathsf{T}}$, and $\ell(L) \approx (0.013506054, 0.005470926)^{\mathsf{T}}\,\mathrm{m}$. Calculations use the full exported coefficients in this fixed chart, shared by learning, integration and stability analysis.

2.2 Complex energy and a power-consistent decoder

The complex Hamiltonian follows by an invertible change of coordinates. Let $x = (X,Y) \in \mathbb{R}^{2n}$, $J = \left(\begin{smallmatrix} 0 & I \\ -I & 0 \end{smallmatrix}\right)$, and

$$\zeta_j = \frac{X_j + \mathrm{i}Y_j}{\sqrt{2}}, \qquad \chi_j = \frac{X_j - \mathrm{i}Y_j}{\sqrt{2}},$$
$$X_j = \frac{\zeta_j + \chi_j}{\sqrt{2}}, \qquad Y_j = \frac{\zeta_j - \chi_j}{\mathrm{i}\sqrt{2}}. \tag{5}$$

On the real phase space $\chi = \bar{\zeta}$. Substitution into (4) gives the explicit complex form

$$h_*^{\mathbb{C}}(\zeta,\chi) = \sum_j \nu_j \zeta_j \chi_j + 1 - \cos\left(\frac{d^{\mathsf{T}}(\zeta+\chi)}{\sqrt{2}}\right) - \tfrac{1}{4}[d^{\mathsf{T}}(\zeta+\chi)]^2. \tag{6}$$

It restricts to the real energy when $\chi = \bar{\zeta}$ and is entire in the independent complex variables (ζ,χ). This extension, including the learned polynomial residual below, supplies the analytic Hamiltonian used in KAM theory. It need not be holomorphic in ζ alone. For a real analytic energy h, the chain rule yields

$$\{\zeta_j, \bar{\zeta}_k\} = -\mathrm{i}\delta_{jk}, \qquad \partial_{\bar{\zeta}_j} = \frac{\partial_{X_j} + \mathrm{i}\partial_{Y_j}}{\sqrt{2}},$$
$$\zeta' = -\mathrm{i}\partial_\chi h^{\mathbb{C}}, \qquad \chi' = \mathrm{i}\partial_\zeta h^{\mathbb{C}}. \tag{7}$$

Magnetic actuation and damping enter through the physical decoder. A CHNN supplies real readouts h_η and $\mathcal{D}_\eta(X,v)$, with physical dissipation divided by $E_0\omega_0$, and

$$X' = v = \partial_Y h_\eta,$$
$$Y' = -\partial_X h_\eta - \partial_v \mathcal{D}_\eta(X,v) + u\,d\cos(d^{\mathsf{T}} X). \tag{8}$$

Here $u = B_y/B_0$, and $\partial_v \mathcal{D}_\eta$ is evaluated before substituting $v = \partial_Y h_\eta$. With $\mathsf{N} = \mathrm{diag}(\nu_j)$, we use $\mathcal{D}_\eta = \widehat{\gamma} v^{\mathsf{T}} \mathsf{N}^{-1} v/2$, $\widehat{\gamma} \geq 0$; for the stated beam, $\gamma = \alpha/\omega_0$. Taking the derivative along the decoded trajectory gives

$$h_\eta' = -v^{\mathsf{T}} \partial_v \mathcal{D}_\eta + u\cos(d^{\mathsf{T}} X) d^{\mathsf{T}} v, \tag{9}$$

with an additional $\partial_\tau h_\eta$ for time-varying parameters. The quadratic Rayleigh readout makes the first term nonpositive. In complex coordinates this decoder becomes

$$\zeta' = -\mathrm{i}\partial_{\bar{\zeta}} h_\eta + \frac{\mathrm{i}}{\sqrt{2}}[-\partial_v \mathcal{D}_\eta + u\,d\cos(d^{\mathsf{T}} X)]. \tag{10}$$

One may package the readouts as $\mathcal{H}_\eta = h_\eta + \mathrm{i}\mathcal{D}_\eta$, but this pair is distinct from the analytic continuation $h_\eta^{\mathbb{C}}$: its dynamics are defined by the power-consistent decoder above. The symplectic scheme acts on the conservative energy with $u = 0$ and $\mathcal{D}_\eta = 0$.

2.3 Derivation of the fourth-order central scheme

For $h = h_\eta$ and $f = J\nabla h$, we derive the specialization of Feng et al.'s classical fourth-order central scheme (Feng et al. 1989, Theorem 24, p. 90; fourth-order formula, p. 95) to the full learned magnetic energy. The generating-function

construction (Hairer et al. 2006) retains all mixed position–momentum terms. Let Δ denote the dimensionless step and $m = (x_- + x_+)/2$ the endpoint midpoint. A central generating function $\mathcal{S}_\Delta$ defines a canonical map through

$$x_+ - x_- = J\nabla\mathcal{S}_\Delta(m). \tag{11}$$

Taking $\mathcal{S}_\Delta = \Delta h$ gives implicit midpoint. To determine the cubic correction, expand the exact trajectory about its temporal midpoint $x_0 = x(t + \Delta/2)$:

$$\begin{aligned} m &= x_0 + \tfrac{1}{8}\Delta^2 f' f(x_0) + O(\Delta^4), \\ x_+ - x_- &= \Delta f(x_0) \\ &\quad + \tfrac{1}{24}\Delta^3\{f''[f,f] + (f')^2 f\}(x_0) + O(\Delta^5). \end{aligned} \tag{12}$$

Since $x_0 = m - \Delta^2 f' f(m)/8 + O(\Delta^4)$, substitution yields

$$\begin{aligned} x_+ - x_- &= \Delta f(m) \\ &\quad + \frac{\Delta^3}{24}\{f''[f,f] - 2(f')^2 f\}(m) + O(\Delta^5). \end{aligned} \tag{13}$$

The coefficient $-2/24 = 1/24 - 1/8$ accounts for the difference between the temporal and endpoint midpoints.

To recover a scalar generator, put $A = \nabla^2 h$. Since $f' = JA$ and $(f')^\mathsf{T} = -AJ$, differentiation gives

$$\begin{aligned} \nabla(f^\mathsf{T} A f) &= h^{(3)}[f,f] - 2AJAf, \\ J\nabla(f^\mathsf{T} A f) &= f''[f,f] - 2(f')^2 f. \end{aligned} \tag{14}$$

Consequently choose

$$\mathcal{S}_\Delta(x) = \Delta h(x) + \frac{\Delta^3}{24} f(x)^\mathsf{T} \nabla^2 h(x) f(x). \tag{15}$$

Equations (11) and (14) give the fourth-order difference scheme

$$\begin{aligned} x_+ &= x_- + g_\Delta(m), \\ g_\Delta &= \Delta f + \frac{\Delta^3}{24}[f''[f,f] - 2(f')^2 f]. \end{aligned} \tag{16}$$

Equivalently, $\zeta_+ - \zeta_- = -\mathrm{i}\partial_{\bar\zeta}\mathcal{S}_\Delta(m)$.

Each step solves $r(x_+) = x_+ - x_- - g_\Delta(m) = 0$. Newton's correction is

$$x_+^{(k+1)} = x_+^{(k)} - \left[I - \tfrac{1}{2} D g_\Delta(m^{(k)})\right]^{-1} r(x_+^{(k)}). \tag{17}$$

The Jacobian uses energy derivatives through order four and is invertible near the identity for sufficiently small Δ. Analytic polynomial and cosine derivatives therefore provide an implementable scheme even when learned momentum terms preclude the physical kinetic–potential split.

2.4 End-to-end complex fully connected Hamiltonian learning

The finite-feature energy in Section 3.2 permits coefficient certificates. A second model learns the hidden representation itself. In the same fixed dimensionless canonical chart, let $\widetilde{z} = (X + \mathrm{i}Y)/(\sqrt{2}s_x)$, with fixed training scale $s_x > 0$, and define

$$\begin{aligned} a_1 &= W_+\widetilde{z} + W_-\overline{\widetilde{z}} + b_1, \\ h_1 &= \tanh(a_1), \qquad a_2 = W_2 h_1 + b_2, \\ h_2^{\mathrm{net}} &= \tanh(a_2), \qquad r_\theta(x) = \mathrm{Re}(w_o^\mathsf{T} h_2^{\mathrm{net}}), \\ H_\theta(x) &= H_{\mathrm{prior}}(x) + s_h[r_\theta(x) - r_\theta(0)]. \end{aligned} \tag{18}$$

Here every hidden weight is trained, $s_h > 0$ is a fixed output scale, and $\tanh$ is the complex hyperbolic tangent, rather than a split activation. The two prior choices are $H_{\mathrm{prior}} = 0$ and h_2 from (4). Complex affine multiplication follows Trabelsi et al. (2018, Section 3.2); the conjugate input allows nonholomorphic dependence on $\widetilde{z}$, including physical quadratic energies. The subtraction fixes the constant gauge only. Even when h_2 is added, the learned residual can change the gradient, Hessian and cubic terms.

Complex $\tanh$ has poles, so trainable weights are parameterized with row constraints. Write the first preactivation as $a_1 = C_X X + C_Y Y + b_1$. On the declared real box $|X_j| \le r_{X,j}$, $|Y_j| \le r_{Y,j}$, enforce, for each row k,

$$\begin{aligned} \sum_j \big(|\operatorname{Im} C_{X,kj}| r_{X,j} + |\operatorname{Im} C_{Y,kj}| r_{Y,j}\big) &< \rho_1 - |\operatorname{Im} b_{1,k}|, \\ \sum_j \big(|\operatorname{Im} W_{2,kj}| + \tan(\rho_1)|\operatorname{Re} W_{2,kj}|\big) &< \rho_2 - |\operatorname{Im} b_{2,k}|. \end{aligned} \tag{19}$$

The implementation uses $\rho_1 = 0.15$, $\rho_2 = 0.6$, both below $\pi/4$. These constraints bound the imaginary preactivations while retaining the real directions for learning. They leave the activation itself unchanged.

Proposition 1. Pole exclusion on a real training box. *Under (19), both hidden layers are well-defined on the declared box. In layer $\ell = 1, 2$, $|\cosh a_\ell| \ge \cos\rho_\ell$, $|\operatorname{Re} h_\ell| \le 1$, and $|\operatorname{Im} h_\ell| \le \tan\rho_\ell$ componentwise. For each fixed finite set of weights, H_θ is real analytic there and has a holomorphic extension to a nonzero complex neighbourhood.*

Proof. For real a, b, $|\cosh(a + \mathrm{i}b)|^2 = \sinh^2 a + \cos^2 b$ and $\tanh(a + \mathrm{i}b) = (\sinh 2a + \mathrm{i}\sin 2b)/(\cosh 2a + \cos 2b)$.

When $|b| < \rho < \pi/4$ these identities give the three bounds. The first row constraint controls $\operatorname{Im} a_1$; its output bounds and the second constraint control $\operatorname{Im} a_2$. The positive pole margin, compact box and analytic composition yield the extension. Supplementary Proposition S13 gives an explicit radius that can subsequently be enclosed with interval arithmetic.

For the joint dissipative model, a position-only real network $2 \to 4 \to 2$ with 22 trainable parameters supplies nonnegative $\gamma_{\theta,j}(X)$. It has separate weights from (18); both parameter sets are optimized in the same objective. The derivative-labelled and native COMSOL window objectives are specified separately. Define

$$\begin{aligned} R_\theta(X) &= \operatorname{diag}\big(\gamma_{\theta,j}(X)/\nu_j\big), \\ \mathcal{D}_\theta(X,v) &= \tfrac{1}{2} v^{\mathsf{T}} R_\theta(X) v, \\ f_\theta(x,u) &= \begin{pmatrix} \partial_Y H_\theta \\ -\partial_X H_\theta - R_\theta(X)\partial_Y H_\theta + u\, d\cos(d^{\mathsf{T}}X) \end{pmatrix}. \end{aligned} \tag{20}$$

The nonnegativity map and training constants are specified with the numerical protocol. This is a diagonal specialization of the positive-semidefinite port-Hamiltonian construction of Zhong et al. (2020, Eq. (3), Section 3.5), with the magnetic port fixed by the beam model. Their original architecture learns the input matrix; fixing it here isolates energy and damping identification. In the derivative-labelled simulator experiment, the loss compares f_θ with the observed state derivatives, and automatic differentiation differentiates H_θ with respect to the original canonical variables, including the input scaling. Conservative data use zero damping and zero transverse input; the derivative-labelled joint branch also uses dissipative data. The COMSOL identification uses trajectory windows and the observed full-beam correction in Section 4.5, with independently initialized networks.

With $v = \partial_Y H_\theta$, the decoder gives the exact continuous power identity

$$\frac{dH_\theta}{d\tau} = -v^{\mathsf{T}} R_\theta(X) v + u\cos(d^{\mathsf{T}}X) d^{\mathsf{T}} v. \tag{21}$$

The derivation differentiates $\mathcal{D}_\theta$ in its independent velocity argument before substituting v. Position derivatives of R_θ do not enter that force. Complex hidden states and readouts therefore generate a real physical vector field. A packaged output $H_\theta + \mathrm{i}\mathcal{D}_\theta$ is a pair of dimensionless quantities decoded by (20), not an instruction to substitute a complex energy into the ordinary real Hamilton equations. The positive power loss constrains the model's structure; its accuracy is tested separately.

A geometry-induced viscous extension. A positive distributed linear-viscosity law and a nonnegative endpoint contribution preserve the same conservative energy and magnetic port. Supplementary Section S3.25 proves uniform dissipative coercivity and nonconstant mass-normalized damping for the frozen planar geometry. A separately generated COMSOL pilot for the distributed-viscosity reference has validated continuous-IVP and nodal observation enclosures in Supplementary Section S3.26. Its quadratic output-class remainder is also bounded. Identification of this nonconstant law requires an adequately excited design and a new learning-error budget.

3 Numerical and approximation analysis

3.1 Symplecticity, order and phase accuracy

For a differentiable local solution of (16), set $C_\Delta = J\mathcal{S}''_\Delta(m)/2$. Differentiating the implicit map gives

$$D\Psi_\Delta = (I - C_\Delta)^{-1}(I + C_\Delta). \tag{22}$$

Since $\mathcal{S}''_\Delta$ is symmetric, $C_\Delta^{\mathsf{T}} J + J C_\Delta = 0$, so this Cayley transform satisfies $(D\Psi_\Delta)^{\mathsf{T}} J D\Psi_\Delta = J$. The odd dependence $\mathcal{S}_{-\Delta} = -\mathcal{S}_\Delta$ also gives $\Psi_{-\Delta} = \Psi_\Delta^{-1}$. The local defect $O(\Delta^5)$ in (13) implies fourth-order convergence on fixed intervals with bounded derivatives and locally unique implicit steps. These classical properties (Feng et al. 1989, Theorems 17–18 and 23–24) hold for the complete fitted energy; the computation below measures the additional error from finite Newton solves.

The linear oscillator makes the phase improvement explicit. For $h = \nu(X^2 + Y^2)/2$, put $a = \nu\Delta + (\nu\Delta)^3/12$. The step is a rotation with eigenvalues $(1 \pm \mathrm{i}a/2)/(1 \mp \mathrm{i}a/2)$ and phase magnitude

$$\theta_\Delta = 2\arctan(a/2) = \nu\Delta - \frac{(\nu\Delta)^5}{120} + O(\Delta^7). \tag{23}$$

Its modulus is one for every real step. Implicit midpoint instead has $\theta_\Delta = \nu\Delta - (\nu\Delta)^3/12 + O(\Delta^5)$. Both preserve the quadratic energy exactly while accumulating different phase errors. For a nonlinear energy, symplecticity preserves canonical structure and does not imply exact energy conservation.

We test this distinction on the prior-informed complex fit from training seed 17 (GC17, defined in Section 3.3), retaining its complete cosine and quartic terms. From initial state $(0.1, 0.06, 0, 0.02)$ we integrate to $\tau = 10$ against a

Table 2. Endpoint errors for the complete fitted GC17 energy. The state error is Euclidean; orders use successive step halvings.

Δ	Midpoint error (order)	Central4 error (order)
0.08	7.1351×10^{-2} (—)	1.8346×10^{-3} (—)
0.04	1.9383×10^{-2} (1.8802)	1.1784×10^{-4} (3.9606)
0.02	4.8966×10^{-3} (1.9849)	7.4149×10^{-6} (3.9902)
0.01	1.2265×10^{-3} (1.9972)	4.6422×10^{-7} (3.9976)

DOP853 reference whose tolerance refinement changes each component by at most 8.32×10^{-14}. Table 2 confirms the predicted orders.

The maximum implicit residual is 2.00×10^{-15} and the maximum symplectic defects are 3.34×10^{-16} for Central4 and 2.24×10^{-16} for midpoint. An independent finite-difference tangent agrees within 1.39×10^{-12}. These results confirm fourth-order convergence and structure preservation at the measured solve accuracy. The comparison uses equal step sizes; Central4 requires higher derivatives. Supplementary Section S9.1 gives the solver tolerances and complete checks.

3.2 Complex features and derivative approximation

Having controlled integration error, we next control the learned energy and its derivatives. The complex form (6) suggests retaining the magnetic cosine and learning a quartic residual. For two modes, let $m_4(x)$ list the 35 degree-four monomials, v_4 contain the coefficients of $-(d^{\mathsf{T}}X)^4/24$, and the 34 columns of U span its orthogonal complement. Define

$$F(x) = \begin{pmatrix} g(d^{\mathsf{T}}X)/\|v_4\|_2 \\ U^{\mathsf{T}} m_4(x) \end{pmatrix},$$
$$h_\eta(x) = h_2(x) + \eta^{\mathsf{T}} F(x) = h_2 + a\, g(d^{\mathsf{T}}X) + m_4(x)^{\mathsf{T}} U c, \tag{24}$$

where $\eta = (a\|v_4\|_2, c)$. This class contains the target at $a = 1, c = 0$ without duplicating its leading quartic direction. Substituting (5) realizes each monomial with complex affine forms and their conjugates; square activations implement products via $uv = [(u+v)^2 - (u-v)^2]/4$. The cosine and these polynomial features give a finite analytic complex network with a real linear readout and allow momentum-dependent residuals.

More generally, increasing the polynomial degree approximates an analytic target and each fixed order of its derivatives on a compact polydisk inside the power-series convergence domain. The fixed class above instead admits a quantitative coefficient bound. This derivative requirement differs from the L^2 comparison for the particular function classes in Zhang et al. (2022, Theorem 2).

For an in-class target $h_* = h_2 + \eta_*^{\mathsf{T}} F$, define $B_r(D) = \max_{|\beta|\le r} \sup_{x\in D} \|\partial^\beta F(x)\|_2$. For conservative derivative labels, let ℓ be the mean over N observations of the squared Euclidean residual of each four-component vector, and ϵ_{rms} the corresponding label-error bound. With $DF \in \mathbb{R}^{35\times 4}$ and $G = N^{-1}\sum_i DF(x_i)DF(x_i)^{\mathsf{T}}$, the triangle inequality gives

$$\begin{aligned} \|h_\eta - h_*\|_{C^r(D)} &\le B_r(D)\|\eta - \eta_*\|_2 \\ &\le \frac{B_r(D)(\sqrt{\ell} + \epsilon_{\mathrm{rms}})}{\sqrt{\lambda_{\min}(G)}}, \end{aligned} \tag{25}$$

provided G is positive definite; the C^r norm is the maximum over scalar coordinate derivatives. Indeed, orthogonality of J equates the vector-field coefficient error to the derivative-feature error, and the least eigenvalue of G bounds its empirical squared norm from below. This model-specific estimate includes nonzero fitting and label errors. A target outside the class requires an additional approximation defect in both the label budget and function error. Section 4 uses these quantities to test KAM conditions.

3.3 Matched identification and long-time prediction

Five paired comparisons separate representation from the mechanical prior. R and C denote real and complex features; G fixes the physical quadratic energy and B fits its 10 coefficients. Thus GR and GC fit 35 real parameters, and BR and BC fit 45, using the fixed chart (3). Within each prior choice, both representations use identical function classes, data and regularizers in common real coordinates.

For each seed in $\{17, 43, 91\}$, the two-mode Galerkin model generates 12 conservative 4 s trajectories with 129 observations each. States are noiseless; vector-field components receive uniform noise in $[-10^{-6}, 10^{-6}]$. Regularized least squares fits the readout, and eight independently initialized 16 s trajectories are held out in full. Supplementary Section S5.1 specifies initial-state sampling, regularization and solver tolerances.

Abbreviating BR and BC as R and C, R−GR and C−GC measure the prior effect, R−C and GR−GC compare representations, and (GR−GC)−(R−C) measures their interaction. Differences are paired within each training seed before computing means and sample standard deviations.

The mean tip RMSE is $2.28 \times 10^{-5}\,\mathrm{mm}$ for BR/BC and $1.20 \times 10^{-5}\,\mathrm{mm}$ for GR/GC. The prior improves two seeds and worsens the third. Representation and interaction

Table 3. Five paired contrasts of tip RMSE, using the same eight held-out trajectories for each of three training seeds. Positive prior contrasts favour G; positive representation contrasts favour C.

Contrast	Mean ± sample SD (mm)
R − GR	$(1.08\pm1.44)10^{-5}$
C − GC	$(1.08\pm1.44)10^{-5}$
R − C	$(-4.24\pm4.84)10^{-13}$
GR − GC	$(0.75\pm1.99)10^{-15}$
(GR − GC) − (R − C)	$(4.25\pm4.82)10^{-13}$

contrasts are smaller than the reference-solver refinement change, 1.92×10^{-8} mm; the matched real and complex fits are numerically equivalent here. The three-seed summaries in Table 3 are descriptive. GC estimates $a = 1.00326 \pm 0.00154$ with nonzero training residuals. Supplementary Table S15 gives slope and physical target-energy errors as well as tip errors. The small errors reflect an in-class target and accurately controlled synthetic data.

A separate 200 s continuation spans 363 first-mode periods and gives GC tip RMSEs of 1.38×10^{-4}–4.03×10^{-4} mm, maximum modal phase error 4.19×10^{-4} rad, and own-energy relative drift below 5.11×10^{-9}. These DOP853 trajectories test long-time prediction, independently of the Central4 test (supplementary Figure S4). Their finite-time agreement leads to the separate invariant-torus question in Section 4.

3.4 Dissipation and independent finite-element comparisons

Driven tests first isolate the role of dissipation. Fitting the Rayleigh coefficient on unforced trajectories gives tip RMSEs of 6.40×10^{-5}–1.37×10^{-4} mm on six held-out forced trajectories; omitting it gives approximately 3.01 mm. The largest work-balance residual with dissipation is 4.41×10^{-17} J. These free-motion tests precede the clearance constraints introduced in Section 5.

Independent COMSOL calculations then quantify reduction error. On an 8 s driven test, exact-cosine predictors with 2, 4 and 8 modes give tip RMSEs of 0.0156168, 0.00407422 and 0.00102484 mm, respectively, versus 0.00005614 mm for all 64 modes. The learned two-mode GC error, 0.0156171–0.0156177 mm, shows that reduction dominates this test. Supplementary Section S5.4 gives the assembly and refinement checks.

Fitting position–velocity increments from separate full-FE COMSOL trajectories gives $a = 1.0223248822$ and test errors 0.14836 and 0.05140 μm, versus 0.24172 and 0.27459 μm for the exact two-mode predictor. The learned residual partly absorbs reduction error (Supplementary Section S5.5). The KAM proof instead uses separate two-mode COMSOL observations, so that its identification error can be bounded directly.

Unknown-parameter and model-mismatch tests further assess the feature choice (Supplementary Section S5.7). For a soft rod with positive field gradient, adding omitted physical features reduces tip RMSE from 182.988 to 9.829 μm; a freely fitted quadratic model gives 7.258 μm. At 0.5% state noise, the corresponding errors are 183.028, 9.757 and 7.216 μm. Thus a physical prior can limit absorption of reduction error. Finally, on six one-second forecasts along a slowly drifting full-FE trajectory, causal adaptation from past states reduces mean error from 387.483 to 9.867 μm. These nonautonomous forecasts have a separate empirical scope from the stationary KAM result.

3.5 Deep-network factorial and joint-learning experiment

The extension uses the same modal matrices, canonical scales, frequencies and magnetic coupling as the finite-feature study. Their frozen arrays were checked for exact equality. Here the training derivatives are generated by the reduced Python model; this experiment does not replace the finite-feature state-only COMSOL identification. The independently initialized deep COMSOL joint fit and its conservative invariant-torus result are described in Section 4.5. To distinguish the two model families, write R_D, C_D, GR_D, GC_D for the deep real/complex networks without/with the quadratic prior. Each has two hidden layers: width 24 and 744 real parameters for the real networks, versus width 16 and 736 real parameters for the complex networks. Both complex groups use the literal activation and row constraints in Section 2.4.

The protocol is frozen before optimization. Nine complete conservative orbits supply 1161 training states and three supply 387 validation states, with clean derivative labels and data seed 17. Network seeds $17, 43, 91$ give three paired runs per group. All groups use 4000 Adam steps followed by 20 L-BFGS calls of at most 10 iterations each; the smallest validation loss selects the checkpoint. Input scaling uses training states only. Output and loss scaling use the full target field for R_D, C_D and its nonlinear residual for GR_D, GC_D. Thus normalized training losses are not comparable across prior choices. Each frozen energy is followed by a two-scalar positive damping fit on six additional orbits, with two orbits for validation. The known magnetic port is retained. Supplementary Section S3.27

Table 4. Deep-model results, shown as the range over three network seeds. E_{nl} is a ratio, not a percentage. Driven-state error is the relative sample L^2 error over 2 s, in percent. The final row is one validation-selected joint run.

	Existing E_{nl}	New E_{nl}	New driven (%)
R_D	691–851	623–856	69.1–77.3
C_D	598–794	848–1006	34.8–61.6
GR_D	.0459–.0773	.00120–.00205	.01098–.01159
GC_D	.0603–.1043	.00241–.01163	.01014–.01090
Joint GC_D	.0550	.001570	.010183

specifies optimization, acceptance criteria and the joint branch.

We evaluate the existing held-out trajectories and a second set of initial conditions generated only after every checkpoint was frozen. The latter uses seed 2026101208, eight conservative trajectories of 1 s and six driven trajectories of 2 s, with the same beam and previously specified drives. Existing field scores use stored samples over 16 s; new field scores use all samples over 1 s. The two sets therefore differ in both initial conditions and sampling window; their error magnitudes are not a before/after training comparison. All comparisons within a test set use the same reference states.

For reference field $f_\star$ and quadratic field $f_2 = J\nabla h_2$, the common nonlinear error is

$$E_{\mathrm{nl}} = \frac{\|f_\theta - f_\star\|_{\mathrm{samples}}}{\|f_\star - f_2\|_{\mathrm{samples}}}, \tag{26}$$

where the sample norm is the Euclidean norm over all states and components. This denominator exposes errors hidden by the dominant quadratic field. Table 4 and Figure 1 report the actual outcomes. Every trajectory completed within the declared box. All six prior-informed runs passed the predefined checks on both test sets; none of the six runs without the prior did. For example, the latter have existing total-field errors below 1% but nonlinear errors between 598 and 851 and driven-state errors of 23.6–54.8%. The prior-informed parameterization and residual scaling are consequential under this optimization budget.

The five paired contrasts are reported in Table 5. For the new nonlinear field error, $GR_D - GC_D$ is negative in every seed: the real model is more accurate. For driven-state error, the complex prior-informed model is slightly better in every seed. Neither observation establishes general complex superiority. In addition to representation, the real/complex comparison includes the complex pole constraints and slightly different parameter counts. Three initialization seeds on one training dataset describe optimization variability, not independent physical replications.

Table 5. Mean of the three seed-paired differences on new initial conditions. Each difference is first evaluated within a seed. Positive simple differences favour the second operand; the interaction is the change in the representation contrast after adding the prior. Driven differences are percentage points. All 15 component differences per metric are supplied.

Contrast	E_{nl}	Driven error
$R_D - GR_D$	755.098	72.8329
$C_D - GC_D$	912.623	51.0881
$R_D - C_D$	−157.529	21.7455
$GR_D - GC_D$	−0.004051	0.0007235
$(GR_D - GC_D) - (R_D - C_D)$	157.525	−21.7448

The joint experiment starts from GC_D seed 43, selected by its conservative validation loss, and optimizes the energy and the 22-parameter damping network together. Both networks' parameter tensors change. It passes both test sets and gives $E_{\mathrm{nl}} = 0.001570$ and 0.010183% driven-state error on the new initial conditions. Because the reference damping is constant, this tests joint fitting and the power-consistent decoder rather than recovery of an unknown spatial damping law. It is one prespecified extension, not a three-seed factorial comparison of joint training.

For analytic-domain verification, the selected and joint complex energies are exported with exact binary64 effective coefficients. Outward-rounded interval calculations certify complex neighbourhoods of the real box with respective radii at least 5.58785×10^{-4} and 6.31070×10^{-4} in the dimensionless maximum norm. The proof and export scope appear in Supplementary Section S3.27. Their complete fourth-derivative tensor errors at the origin, relative in Frobenius norm to the analytic Python reference energy, are 5.94% and 9.39%: joint fitting improves some prediction errors while worsening this derivative diagnostic. Small nonzero linear and cubic terms are also present. Consequently a new normal-form or a posteriori torus computation must use the actual learned function; the finite-feature KAM result below does not certify these derivative-labelled fits. The COMSOL-trained seed-43 network is a distinct frozen function despite the shared architecture and initialization seed.

4 Connection to KAM nondegeneracy and invariant tori

4.1 From state observations to analytic error

The results in Sections 4.1–4.4 concern the finite-feature Hamiltonian fitted from two-mode COMSOL states. Supplementary Section S3.6.4 gives a conditional route for an independently trained deep energy, including the new domain, derivative and a posteriori checks that such a transfer

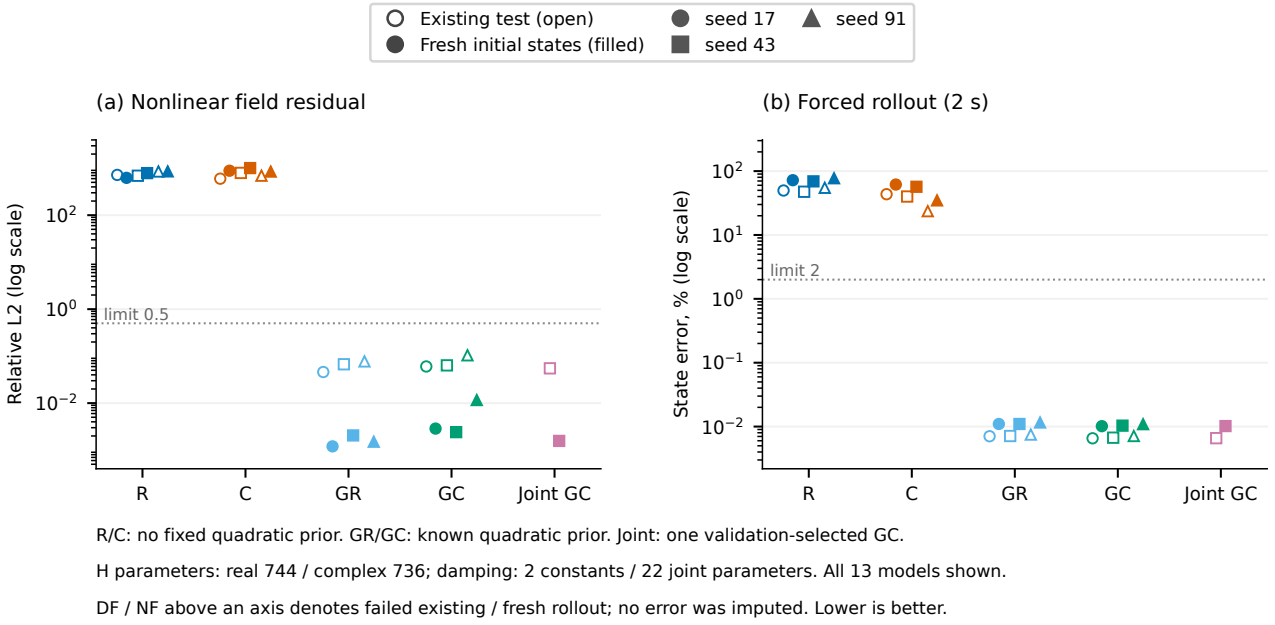


Figure 1. Frozen deep-network comparisons. R, C, GR and GC in this figure denote the deep groups R_D, C_D, GR_D, GC_D. Every seed is retained, including unsuccessful fits. New initial conditions are evaluated after checkpoint freezing. These are clean-label reduced-model experiments with a known magnetic port.

requires. Section 4.5 gives an independent conservative existence result for the COMSOL-trained deep Hamiltonian. Long-time prediction tests the fitted dynamics along selected trajectories. To establish invariant motion, we also need error bounds on a neighborhood of those trajectories. Following the learning-to-KAM connection of Chen et al. (2022, Theorem 11), we derive these bounds for the analytic readout (24), $h_\eta = h_2 + \eta^{\mathsf{T}} F$, whose quadratic part is fixed and whose full-cosine target is $h_\star = h_{\eta_\star}$. We write $\widehat{h}$ for the fitted function, including coefficient-export rounding.

For independently sampled gradient labels, our uniform learning bound in Supplementary Theorem S3 connects nonzero loss to a C^ℓ error under explicit sampling and regularity assumptions. Supplementary Corollary S4 combines it with Bounemoura (2020, Theorem A). The COMSOL observations used here are correlated states along trajectories, so we instead derive a deterministic bound from their integrated equations of motion.

COMSOL supplies 17 position–velocity observations on each of $m = 64$ conservative two-mode trajectories. Let $c_j : [0, T] \to \mathbb{R}^4$ be their degree-eight fitted curves, with endpoints fixed to the observations, and let $Lx = J\nabla h_2(x)$. Define

$$\begin{aligned} D_j &= T^{-1} \int_0^T J\nabla F(c_j(t))\, dt, \\ y_j &= T^{-1} \left[c_j(T) - c_j(0) - \int_0^T L c_j(t)\, dt \right], \\ G_I &= (4m)^{-1} \sum_j D_j^{\mathsf{T}} D_j. \end{aligned} \tag{27}$$

These labels require states and feature integrals, without acceleration reconstruction. Let r_I be the RMS residual $D_j\widehat{\eta} - y_j$ over the $4m$ scalar observations, and let ϵ_I bound the RMS integrated physical defect $e_j = T^{-1}\int_0^T (\dot{c}_j - J\nabla h_\star(c_j))\, dt$.

Proposition 2. Integral observations bound the learned energy. *If* $\lambda_{\min}(G_I) \geq \underline{\lambda} > 0$, *then*

$$\begin{aligned} \|\widehat{\eta} - \eta_\star\|_2 &\leq b_I := \frac{r_I + \epsilon_I}{\sqrt{\underline{\lambda}}}, \\ \|\widehat{h} - h_\star\|_r &\leq b_I B_F(r) + \delta_{h,\mathrm{dec}}(r). \end{aligned} \tag{28}$$

Here $\|\cdot\|_r$ *is the supremum norm on the complex polydisc* $|x|_\infty < r$, $B_F(r)^2 = \sum_j \|F_j\|_r^2$, *and* $\delta_{h,\mathrm{dec}}$ *bounds coefficient-export rounding.*

Proof. Endpoint matching gives $y_j - D_j\eta_\star = e_j$. The triangle inequality in the normalized stacked observation norm bounds $\sqrt{\underline{\lambda}}\|\widehat{\eta} - \eta_\star\|_2$ by $r_I + \epsilon_I$. Cauchy–Schwarz in feature space then gives the analytic bound.

Table 6. Learning error and twist margin for the COMSOL two-mode fit. All displayed bounds include coefficient rounding and are rounded outward.

Quantity	Bound
Training MSE	$[3.6030, 3.6031]\,10^{-23}$
r_I, ϵ_I	$\le 6.003\,10^{-12}$, $6.884\,10^{-12}$
$\lambda_{\min}(G_I)$	$\ge 2.930\,10^{-7}$
b_I	$\le 2.381\,10^{-8}$
Twist error	$\le 1.038\,10^{-7}$
$\sigma_{\min}(A_\star)$	$\ge 4.903\,10^{-4}$
$\|\widehat{h} - h_\star\|_{0.3}$	$\le 1.331\,10^{-9}$

The physical defect is evaluated against the declared target; it is additional information beyond the training loss. Interval integration bounds this defect and retains the full sine remainder and coefficient rounding. Cauchy's estimate then bounds every derivative on a smaller polydisc. Supplementary Section S4.5 gives these estimates and the independent trajectory-error bound.

4.2 The magnetic nonlinearity preserves twist after learning

The analytic error bound becomes useful for KAM theory because the magnetic potential supplies a nondegenerate frequency map. For $h_\star = h_2 + g(d^{\mathsf{T}}X)$, with $g(u) = 1 - \cos u - u^2/2$, its quartic part is $-(d^{\mathsf{T}}X)^4/24$. Averaging over the modal angles with $I_j = (X_j^2 + Y_j^2)/2$ gives the action Hessian

$$A_\star = -\frac{1}{8}\begin{pmatrix} d_1^4 & 2d_1^2 d_2^2 \\ 2d_1^2 d_2^2 & d_2^4 \end{pmatrix}, \qquad \det A_\star = -\frac{3}{64} d_1^4 d_2^4 \neq 0. \tag{29}$$

Indeed, $\langle X_j^4 \rangle = 3I_j^2/2$ and $\langle X_1^2 X_2^2 \rangle = I_1 I_2$, while odd terms average to zero. Both modal tip rotations are nonzero, so the twist is nonsingular, although indefinite.

Learning preserves this property if its quartic error is smaller than the target's least singular value. Let B map the readout coefficients to quartic coefficients and let δ_{dec} bound rounding in that map. Angular averaging yields

$$\|A_{\widehat{h}} - A_\star\|_2 \le \sqrt{19}\big(\|B\|_2 b_I + \delta_{\mathrm{dec}}\big) < \sigma_{\min}(A_\star). \tag{30}$$

The factor $\sqrt{19}$ bounds the quartic-to-twist operator, including momentum-dependent terms. Table 6 establishes the strict inequality for the Hamiltonian learned from the COMSOL states.

The complete magnetic potential also contributes a sixth-order analytic tail bounded by $(Dr)^6 \cosh(Dr)/720$, where $D = \|d\|_1$. Together with fourth-order nonresonance, the twist bound gives a local normal form $\nu \cdot I + \frac{1}{2} I^{\mathsf{T}} A_{\widehat{h}} I + O(r^6)$. Our remainder estimates in Supplementary Sections S4.1–S4.2 therefore yield positive-measure KAM families near equilibrium. To establish persistence at a specified physical amplitude, we next apply an a posteriori theorem to the same fitted function.

4.3 An invariant torus at finite physical amplitude

The existence theorem used here is Haro and Luque (2019, Theorem 3.1), with the constants of arXiv:1711.05515v1, Appendix B. Our contribution in the following result is the construction and analytic bounds for the learned magnetic model.

Theorem 3. Finite-amplitude persistence for the learned magnetic model. *The fitted Hamiltonian in Table 6, specified by its exact coefficients in Supplementary Section S4.5, admits a real analytic embedded Lagrangian torus $K_\infty : \mathbb{T}^2 \to \mathbb{R}^4$ satisfying*

$$J\nabla\widehat{h}(K_\infty) = DK_\infty \varpi. \tag{31}$$

The exact frequency ϖ is specified in Supplementary Eqs. (S302)–(S303); angles have period one and time is dimensionless. The torus is within $2.0894\,10^{-41}$ of the computed Fourier embedding on the complex strip of width $1/80$. Its beam reconstruction attains a tip displacement of at least 1.8977904 mm and has centreline displacement at most 1.8977962 mm at every phase.

Proof. The bounded continued-fraction coefficients of the frequency ratio give $|k \cdot \varpi| \ge (9/1750)|k|_1^{-1}$ for every nonzero integer k, and hence the weaker Diophantine exponent 3 used in the theorem. On the strip $\rho = 1/40$, 512-bit ball-arithmetic evaluation of the Fourier candidate, including interpolation tails, bounds the analytic invariance residual by $2.7310\,10^{-77}$ and its inverse averaged torsion by 80526. The geometric and derivative estimates in Supplementary Section S4.3 give a residual threshold greater than $8.6086\,10^{-76}$, with smallness quotient below $0.031724 < 1$. The cited theorem then supplies the exact torus and correction bound. Fourier bounds and the Hermite-element convex-hull property give the displacement estimates.

Thus nonzero learning loss, preserved twist and finite-amplitude persistence are established for one fitted Hamiltonian. The result uses the independently simulated two-mode state observations; the full-beam comparisons in Section 3 quantify a separate reduction error. Section 5 connects this conservative motion to the controlled full beam.

4.4 From one torus to a family of quasiperiodic motions

The nondegeneracy bound survives the a posteriori iteration: $\|\langle T_\infty\rangle^{-1}\| \le 2\,10^5$ for the exact torus's averaged torsion. In an analytic symplectic tube, solving one quadratic homological equation therefore gives

$$\widehat{h}\circ\Psi = e + \varpi\cdot I + \tfrac{1}{2} I^{\mathsf{T}}\overline{A} I + O(|I|^3), \qquad \det\overline{A}\neq 0. \tag{32}$$

Our construction identifies $\overline{A}$ with the averaged torsion and retains an analytic remainder (Supplementary Corollary S63). The nonresonant frequency, nondegenerate quadratic part and analytic remainder meet the hypotheses of Bounemoura (2016, Theorem 6, $\alpha = 1$, arXiv:1412.0509v1). That theorem gives positive-measure families of analytic Lagrangian invariant tori around K_∞, with relative density tending to one as the transverse action box shrinks. The same argument applies to all seven fitted or target Hamiltonians considered in the supplement.

A numerical fraction additionally depends on the chosen initial-state domain. For $U \subset \mathbb{R}^4$ of positive Liouville volume, define $f_{\mathrm{KAM}}(U) = \mu_L(U\cap\mathcal{K}_h)/\mu_L(U)$, where $\mathcal{K}_h$ is the union of analytic Diophantine Lagrangian tori. Supplementary Section S11 applies Chierchia and Koudjinan (2021, Theorems 1 and 3) to an explicit tube; its very small transverse width limits its use as an operating tolerance. For the broader action domain $I_1, I_2 \in [0.0032, 0.0072]$, independent uniform sampling of actions and phases gives $1024/1024$ finite-time quasiperiodicity candidates over 8192 dimensionless time units (717.5 s). At step $1/64$, all pass the eight-digit test on consecutive windows of length 2048 and 4096; six- and ten-digit tests agree. Supplementary Section S5.6 gives the weighted Birkhoff method, uncertainty and refinement checks. This broader-domain diagnostic complements the local family theorem without supplying an infinite-time volume bound. The magnetic realization below carries the family's low-mode parameter measure into the damped plant; this measure is distinct from volume in the full 128-dimensional state space.

4.5 Weak-form deep identification and conservative persistence

The state-only deep identification experiment uses position–velocity windows from the fixed-mass, geometrically linear COMSOL beam, expressed in the two-mode canonical chart. The complex Hamiltonian has 736 real parameters and a separate 22-parameter positive damping head. Six training trajectories supply $M = 1494$ windows; two complete validation trajectories select each checkpoint. Each fit uses 3000 Adam steps with batches of 256 windows, cosine learning-rate decay from 0.003 to 0.00003, and joint parameter-gradient norm clipping at 10. The initial parameters and checkpoints every 100 steps are evaluated on all validation windows. The best validation checkpoint initializes 20 full-batch L-BFGS calls, each with at most 10 iterations, learning rate 0.5 and a strong-Wolfe line search. The final parameters minimize the recorded validation objective over the initial, Adam and L-BFGS candidates. The mathematical error bounds and conservative certification apply to this selected function without an optimization-convergence assumption. Networks are initialized independently of the derivative-labelled factorial experiment in Section 3.5. Test trajectories were previously inspected during model development; their reuse provides benchmark evaluation rather than blinded validation.

For this subsection write $(q,p) = (X,Y)$ and $N_\nu = \mathrm{diag}(\nu_1,\nu_2)$; time is $\tau = \omega_0 t$. Let $H_\theta = H_2 + h_\theta$, where $H_2 = (q^{\mathsf{T}}N_\nu q + p^{\mathsf{T}}N_\nu p)/2$. The damping matrix is $D_\theta = R_\theta$ from (20). On five equally spaced nodes let Q denote the normalized Boole average with weights $(7,32,12,32,7)/90$. The measured momentum increment, divided by the dimensionless window length h, is $\Delta\widehat{p}/h$. The joint identification objective is defined by

$$\begin{aligned} F_{\theta,p} &= -H_{\theta,q} - D_\theta H_{\theta,p} + C_{\mathrm{obs}},\\ \widehat{w}_i &= Q_i F_{\theta,p} - \Delta\widehat{p}_i/h, \qquad k_\theta = \partial_p h_\theta,\\ L_{\mathrm{mom}} &= \frac{1}{2M}\sum_{i,j}(\widehat{w}_{ij}/s_j)^2,\\ L_{\mathrm{kin}} &= \frac{1}{10M}\sum_{i,\ell,j}(k_{\theta,j}(\widehat{x}_{i\ell})/s_j)^2,\\ L_{\mathrm{joint}} &= L_{\mathrm{mom}} + 0.1 L_{\mathrm{kin}}. \end{aligned} \tag{33}$$

Here $j = 1,2$, $\ell = 0,\dots,4$, and $s_j > 0$ are fixed training-only conditioning scales. Initial damping recentering cancels from $\widehat{w}_i$. The kinematic term penalizes deviation from the fixed velocity chart $H_{2,p} = N_\nu p$; it is not a measured four-component conservative gradient loss.

The observed forcing also retains the discarded-mode tip angle. Writing θ_f for the full-beam tip angle and $\theta_\ell = d^{\mathsf{T}}q$, it is

$$\begin{aligned} C_{\mathrm{obs}} = d\,[&\theta_f - \sin\theta_f - \theta_\ell + \sin\theta_\ell\\ &+ u\cos\theta_f]. \end{aligned} \tag{34}$$

Thus the training field differs from the closed two-mode port $ud\cos\theta_\ell$ in (20). The difference must be

retained or bounded when passing between them. The power identity of the closed decoder alone does not establish equality of these two fields. Supplementary Proposition S6 derives the observed port directly from the fixed-mass full-beam equation: its full-angle term cancels the discarded-mode magnetic contribution exactly. For the literal saved matrices, Supplementary Section S3.3 retains the matrix and coordinate rounding defects. An input-work estimate, followed by interval matrix verification, bounds the momentum defect by $9.504461\,10^{-9}$ in Euclidean norm for the entire 1 s duration of each of the eight training and validation inputs. This bound concerns the continuous native-matrix ODE with the saved initial vectors. Numerical time-integration and observation errors require separate bounds.

Separating conservative and dissipative errors. Suppose a reference trajectory in the same state coordinates obeys

$$q' = H_{*,p} + r_q, \qquad p' = -H_{*,q} - D_* H_{*,p} + C_* + r_{\mathrm{phys}}. \tag{35}$$

Here r_q records a coordinate defect, with $r_q = 0$ in the ideal canonical projection. The following separation uses the momentum equation and remains valid for the literal saved chart. Set $e_p = F_{\theta,p} - p'$, $\Delta H = H_\theta - H_*$, $\Delta D = D_\theta - D_*$ and $\Delta C = C_{\mathrm{obs}} - C_*$. Subtracting the reference and learned equations gives

$$\begin{aligned} \Delta H_p &= k_\theta + (N_\nu p - H_{*,p}), \\ \Delta H_q &= -e_p - D_\theta \Delta H_p - \Delta D H_{*,p} + \Delta C - r_{\mathrm{phys}}. \end{aligned} \tag{36}$$

In Euclidean norm, suppose the respective bounds are $\|k_\theta\| \le K$, $\|N_\nu p - H_{*,p}\| \le \eta_{\mathrm{chart}}$, $\|e_p\| \le E_p$, $\|D_\theta\| \le D_{\max}$, $\|\Delta D\| \le \eta_D$, $\|H_{*,p}\| \le V$, $\|\Delta C\| \le \eta_C$ and $\|r_{\mathrm{phys}}\| \le \eta_{\mathrm{phys}}$. Then, with $E_v = K + \eta_{\mathrm{chart}}$,

$$\begin{aligned} E_q &= E_p + D_{\max} E_v + \eta_D V + \eta_C + \eta_{\mathrm{phys}}, \\ \|\nabla \Delta H\|_2 &\le (E_v^2 + E_q^2)^{1/2}. \end{aligned} \tag{37}$$

Supplementary Propositions S8 and S9 establish the weak-to-pointwise estimate and conservative error bound, including observation, quadrature and regularity terms. An analytic nonidentifiability example shows that both loss terms can vanish while the conservative gradient changes through compensation by the damping. Positivity alone is therefore insufficient for unique energy identification.

For the complex seed-43 COMSOL fit, the finite-sample inequalities evaluated at the fitted losses give

$$\begin{aligned} \max_{i,\ell} \|k_\theta(\widehat{x}_{i\ell})\|_2 &\le s_{\max}\sqrt{10 M L_{\mathrm{kin}}} \simeq 4.0254\,10^{-7}, \\ \max_i \|\widehat{w}_i\|_2 &\le s_{\max}\sqrt{2 M L_{\mathrm{mom}}} \simeq 5.9879\,10^{-7}. \end{aligned} \tag{38}$$

The formulas are exact finite-sample inequalities; the displayed approximations are not outward-rounded certificates. Neither bound is a full-domain error estimate. In particular, the 1494 windows from six fixed trajectories cannot be inserted as 1494 independent state samples into a statistical learning theorem.

For the nominal physical damping coefficient α_{mass}, interval evaluation of the fitted damping head on $[-0.091, 0.091]^4$ yields the independent error bound

$$\begin{aligned} D_* &= \mathrm{diag}\big(\alpha_{\mathrm{mass}}/(\omega_0 \nu_j)\big), \\ \sup \|D_\theta - D_*\|_2 &\le 2.940580\,10^{-7}. \end{aligned} \tag{39}$$

This is an outward-rounded bound obtained by exact rational subtraction from certified interval endpoints. Its reference is the specified nominal damping law; it is not an estimate of unknown physical damping from the loss. Application of (37) requires all remaining error terms on the same domain.

Certified local identification of the nominal energy. The known fixed-mass nominal target also permits direct deterministic verification of the frozen function. With $H_* = H_2 + 1 - \cos(d^{\mathsf{T}} q) - (d^{\mathsf{T}} q)^2/2$, interval Taylor bounds on 65 536 cells covering $[-0.091, 0.091]^4$ establish

$$\begin{aligned} \sup |H_\theta - H_*| &\le 2.137046\,10^{-8}, \\ \sup \|\nabla (H_\theta - H_*)\|_\infty &\le 3.937778\,10^{-6}. \end{aligned} \tag{40}$$

Both bounds are strictly smaller than the exact errors of H_2 on the same box, respectively $2.566601\,10^{-7}$ and $6.631067\,10^{-6}$ to the displayed precision. The common quadratic prior is cancelled before evaluation, so the comparison resolves nonlinear energy rather than agreement with that prior alone. The proof and complete norm conventions are given in Supplementary Section S3.5.

The result is local. At $q = (-0.2, -0.2)$, $p = (0, 0)$, a certified point error exceeds $6.813327\,10^{-5}$, or 96.83% of the quadratic-model worst-case nonlinear field error on $[-0.2, 0.2]^4$. Thus an architecture that is pole-free on a larger box need not identify the nonlinearity accurately throughout that box. The maximum absolute training-state coordinate is approximately 0.19161; local certification does not cover its entire enclosing box, and the witness point need not belong to a recorded trajectory.

Analytic residual control and invariant motion. Supplementary Section S3.7 proves an explicit connection from a valid real-domain energy error ε to an analytic torus residual. In this paragraph the vector norm is the maximum-component norm, and $\|\cdot\|_\rho$ is its supremum on the complex angle strip. A bound B on the pure second derivatives of the energy error and an interior coordinate margin r first give

$$b_r(\varepsilon, B) = \inf_{0<a\le r} (\varepsilon/a + Ba/2). \tag{41}$$

For a candidate K_0 with independently bounded reference residual δ_*, and an independently certified outer-strip residual bound M_R, the residual on $0 < \rho < R$ satisfies

$$\begin{aligned} E_\theta &= J\nabla H_\theta(K_0) - DK_0\omega, \\ \|E_\theta\|_\rho &\le [\min\{M_R, \delta_* + b_r\}]^{1-\rho/R} M_R^{\rho/R}. \end{aligned} \tag{42}$$

The theorem's domain, Diophantine and torsion conditions must also hold. Equation (42) connects a real energy estimate of the type considered by Chen et al. (2022) to the analytic norm required by the a posteriori theorem. The trajectory sampling scheme does not meet the iid state-sampling assumption of that statistical result. A loss-based deterministic application instead requires bounds for all terms in (37), followed by a valid domain-coverage and analytic-extension estimate. Alternatively, the direct frozen-function comparison in (40) supplies a real-domain bound without sampling assumptions. Supplementary Proposition S12 extends that comparison to the candidate's complex tube, giving an error bound of $4.014626\,10^{-5}$. This analytic comparison does not meet the existing certificate's residual budget for direct transfer of the same candidate to the nominal physical Hamiltonian. A point enclosure in Supplementary Section S3.5.3 also proves a positive gradient discrepancy greater than $9.65618\,10^{-9}$ on that candidate, which cannot be removed by increasing the validation sample. The independent nominal construction below addresses a prescribed-frequency comparison without requiring this transfer inequality.

For the fitted complex seed-43 Hamiltonian, a computer-assisted application of Haro and Luque (2019, Theorem 3.1) establishes an analytic embedded invariant two-torus of the conservative vector field $J\nabla H_\theta$. The verified function is specified by its exact binary64 effective coefficients. At candidate amplitude 0.04 and strip width $\rho = 1/160$, the a posteriori smallness quotient is bounded above by $4.888382\,10^{-17} < 1$; the resulting invariant embedding is analytic on the strip of width $1/320$. The frequency ratio is an exact quadratic irrational, and an all-integer Diophantine bound is verified for the corresponding frequency vector. The calculation independently checks the complex domain, derivatives, frame inverses, torsion, residual and embedding corrections for this Hamiltonian. The reproducibility index identifies the exact function and the associated verification inputs.

Nondegeneracy of the learned and nominal conservative energies. The complete fourth-order normal form can also be verified directly for the frozen network, including its shifted equilibrium and cubic elimination term. For $H_\lambda = (1-\lambda)H_\theta + \lambda H_*$, interval calculations over 32 parameter intervals prove a unique elliptic equilibrium near the origin, nonresonance through order four, and

$$\det A_\lambda < -7.2544\,10^{-7}, \quad \sigma_{\min}(A_\lambda) > 3.7525\,10^{-4} \tag{43}$$

for every $\lambda \in [0,1]$. The normal-form and parameter-KAM argument in Supplementary Section S3.8 consequently establishes positive-measure local families of invariant two-tori for each interpolating energy. The calculation evaluates the actual two-layer network; it does not replace its Taylor coefficients by those of the nominal magnetic potential. This local result does not supply a finite-amplitude threshold or identify tori at a common frequency.

A physical invariant torus at the learned frequency. A separately refined Fourier embedding for H_* passes the a posteriori theorem at exactly the certified deep-model frequency. The exact rational first frequency and quadratic-irrational ratio are shared; the physical target retains its full cosine and the same nominal coefficients. The analytic residual is below $2.585\,10^{-109}$, including the interpolation tail, and the smallness quotient is below $7.713\,10^{-20} < 1$. Supplementary Section S3.9 gives the frame, torsion, derivative and embedding checks, repeated at 512 and 640 bits. Thus both specified conservative models possess embedded invariant tori at this same frequency. Their amplitudes and embeddings differ; this assertion does not require the learned embedding itself to solve the nominal equations. With phases paired, the canonical-state distance between the two certified motions is below 0.005041552 for all time. At phase zero the first-coordinate difference exceeds 0.005040448, approximately 12.6% of the deep reference amplitude 0.04. Thus same-frequency persistence does not establish accurate amplitude identification or a same-initial-state prediction bound.

These conservative existence results do not establish identification accuracy, statistical generalization or persistence under the learned damping and magnetic forcing. The full-beam clearance bound in Section 5 concerns the separately

identified finite-feature energy and its magnetic realization; it is not implied by this deep-model torus result.

For configuration-dependent inertia, Supplementary Section S3.10 derives the canonical reduction and complete cubic and quartic kinetic contributions. Supplementary Section S3.11 instantiates them for a planar inextensible Kirchhoff restriction with rotary inertia, using the frozen Hermite basis. Exact polynomial integration supplies its mass two-jet at the exact equilibrium; interval bounds prove analytic kinetic invertibility and $\det A_K \in [-8.580864, -8.580863]\,10^{-4}$. The resulting conservative model has positive-measure local KAM families. Moreover, 89 closed parameter intervals certify the full interpolation $(1-\lambda)H_\theta + \lambda h_K$: its fourth-order invariant remains nondegenerate for every $\lambda \in [0,1]$, with $\sigma_{\min} > 6.6235\,10^{-7}$. This establishes local invariant families throughout the connection to the actual trained function. Its finite-amplitude centreline and canonical momentum differ from the small-slope training model. At the common canonical state $(0,0,0,0.04)$, a certified conservative velocity discrepancy from the frozen network exceeds $1.1136795\,10^{-4}$, so the earlier fixed-mass identification bound cannot be transferred to this rod. On $[-0.04, 0.04]^4$, a direct comparison instead gives $\|\nabla(H_\theta - h_K)\|_\infty < 2.195079\,10^{-4}$. This calculation supplies a concrete variable-inertia existence result; matching COMSOL labels and learned-model certification for that specialization remain separate requirements.

The structured extension in Supplementary Section S3.12 retains this kinetic metric and learns a complex-network potential correction that vanishes through order five. Every fixed admissible parameter vector preserves the certified fourth-order invariant and its qualitative local KAM conclusion. Euler–Lagrange window residuals give the corresponding position–velocity data interface. For constant mass-proportional damping, an explicit time-dependent Birkhoff representation and a matched second-order step retain the conformal symplectic structure of the variable-inertia equations. These components specify the mechanical architecture used in the joint fit below. Its same-chart power and safety results have their own single-port assumptions; the earlier two-port feedback guarantees are not transferred.

For this mechanical architecture, Supplementary Theorem S37 specifies a regularized full-batch training rule with backtracking. Compact optimization level sets yield the required parameter-gradient regularity, finite line searches and convergence of gradient norms to zero in exact arithmetic. Every finite iterate retains the same qualitative conservative KAM conclusion; compactness also yields a common sufficiently small existence scale, without a computed radius or a common torus. This implements a smooth-descent argument in the spirit of Zhang et al. (2022, Theorem 3), with explicit algorithmic assumptions; it supplies neither global fitting accuracy nor the iid statistical premises of the separate learning-to-KAM result. The numerical mechanical evaluators require an additional coefficient check. Supplementary Section S3.14 verifies the training buffers and two interpretations of the exported COMSOL constants separately. Their exact rational kinetic jets and interval normal forms retain ellipticity, fourth-order nonresonance and a nonzero twist determinant. Thus the local conservative conclusion also holds for these specified quadrature functions, without identifying them with the continuous integral or certifying time-solver accuracy.

Joint identification in a matched planar chart. The variable-inertia architecture is trained on fourteen native COMSOL reduced-model trajectories, with four trajectories opened only after optimization. An initial prescribed-drag experiment (Supplementary Section S3.15) is extended by fitting the nonnegative scalar drag together with the 252 real potential coordinates. A training-only eight-column weak fit initializes the leading coefficients; scalar variable projection and 400 full-batch Armijo steps then minimize the same joint window loss. The fitted drag is 0.035000000000117, and pooled held-out hidden-force relative RMS is 0.009085%. This improvement includes the linear initialization and does not establish an advantage of depth or complex arithmetic. Direct ball-arithmetic certification of the frozen raw-parameter function on $|q_i| \le 0.1$ gives potential and conservative-force errors below $2.772\,10^{-13}$ and $1.269\,10^{-11}$, respectively. These are benchmark function bounds, with the specified shared metric, rather than COMSOL solver or continuum-reduction error bounds. Supplementary Sections S3.16 and S3.17 give the certificates and matched descent theorem. For the conservative part of this same frozen fit, the full variable-inertia Hamiltonian satisfies a finite-amplitude a posteriori KAM theorem. A rational Fourier candidate with linear-reference amplitude approximately 0.04 has whole-strip defect below $6.685\,10^{-46}$; the exact smallness quotient is below $9.918\,10^{-9}$. Its invariant embedded torus lies within $8.062\,10^{-28}$ of that candidate on the final analytic strip. This conservative conclusion does not extend unchanged to positive-damping motion or establish iid generalization.

A separately learned complex dissipation factor. With this energy frozen, a second two-hidden-layer complex network learns a configuration-dependent PSD factor from the same fourteen training trajectories. All 42 real and

imaginary coordinates take part in the fixed weak objective. The matched scaled Gauss–Newton and Armijo analysis, whole-box dissipative-field error bound, four held-out evaluations and magnetic work-filter checks are given in Supplementary Section S3.18. The same-chart field error is below $1.269\,10^{-11} + 3.236\,10^{-9} P_{\max}$ on $|q_i| \le 0.1$, $|P_i| \le P_{\max}$. The head reaches the scalar reference's weak-residual scale, with one held-out trajectory worsening. The constant-damping benchmark does not establish nontrivial state-dependent damping or a complex-network advantage.

A posteriori interpretation of the native observations. Supplementary Section S3.19 validates all eighteen native trajectories against the continuous reference IVP, using their original time records and a fixed canonical reconstruction. Whole-interval defect bounds cover all $14\,400$ time intervals; the maximum weighted trajectory error is below $2.114\,10^{-13}$. Native momentum and the momentum reconstructed from the actual training velocity differ by less than $3.185\,10^{-16}$ at all $14\,418$ nodes. The same frozen Hamiltonian and complex head are then evaluated over these tubes to enclose their continuous weak residuals and their total discrepancy from the original cached objective. Under the original normalization, the continuous training weak loss is approximately $2.13219\,10^{-10}$, compared with $2.47987\,10^{-10}$ for the same-energy scalar checkpoint. Two excluded trajectories improve and two worsen. These continuous quantities differ from the original discrete losses. Native momentum and derivative fields are used only for this post-training validation. These statements concern the specified finite-Gauss model and do not certify a continuum reduction or iid generalization.

Conservative accuracy inferred from continuous windows. For the specified reference benchmark, Supplementary Section S3.21 separates seven conservative coefficients and three constant normalized-damping components for the same final output. The continuous 140×10 design has a certified scaled singular value above 0.2880896913927310. Bounding the remaining configuration dependence gives a whole-box conservative-gradient error below $6.967816488\,10^{-11}$ on $|q_i| \le 0.1$, inferred from the continuous weak residual and the declared target-tail class. On the certified complex rectangle, with imaginary half-width 0.001, the corresponding bound is below $7.370756945\,10^{-11}$. These estimates retain the learned damping variation and do not establish statistical generalization. For the same declared finite-Gauss conservative learned/full-cosine pair, the continuous-data discrepancy also controls uniform derivatives and average-torsion nondegeneracy (Supplementary Section S3.22). An explicit third-order parameter response gives a whole-strip residual below $1.205897\,10^{-41}$, against a sufficient threshold above $5.824745\,10^{-38}$. All eight scalar conditions pass, proving that every real point of this Hamiltonian homotopy has an embedded invariant torus at the same exact prescribed frequency (Supplementary Section S3.23). This is a deterministic certificate for the declared pair; it does not prove joint analytic dependence of the exact tori on the parameter or robustness over a mixed coefficient-error ball.

Effective configuration-dependent damping coordinates. Supplementary Section S3.24 extends the same continuous-window design to seven conservative and eighteen quadratic damping coordinates. Its scaled singular value is above 0.01326602753076006, including a positive lower bound for the fifteen nonconstant coordinates after eliminating conservative and constant-damping nuisance terms. The frozen head's certified quadratic Taylor remainder and the declared reference-tail class give a whole-box normalized damping-matrix error below $1.020023\,10^{-7}$. The joint nonconstant zero block remains inside the certified error ball, so these data do not establish a nonconstant true damping law. The preceding, sharper conservative certificate remains the input to the same-frequency KAM result.

Proposition S59 transfers the certified damping-output error to a true dissipation bound $\gamma_{\mathrm{cert}} > 0.034999897$; the corresponding upper output bound also controls its port-derivative contribution. Together these bounds close the dissipation interface to the sampled safety theorem of Supplementary Section S3.20, provided Theorem S53 is implemented with the new reserve cap C_{cert} and the stated reference-model hypotheses.

Sampled magnetic power with deterministic errors. Supplementary Section S3.20 extends the same single-port energy filter to held, imperfect port samples and bounded execution errors over a prescribed finite horizon. An exact held-data work ledger and an endpoint-tightened command constraint preserve the energy and clearance barriers. A uniform guard-domain certificate gives nonzero error allowances for the unchanged reference task. Its less conservative reserve requires the stated true damping lower bound; positivity of the learned head alone does not establish it. The example requires a hold age below approximately $88\,\mu$s and is not a measured hardware specification or a tracking guarantee.

5 Guidance for magnetic actuation and clearance

The finite-feature invariant torus established in Theorem 3 specifies the bounded conservative motion considered here. We now derive the magnetic commands that sustain it in the damped beam and bound the response of all remaining modes. This construction gives an operating envelope for recurrent motion. A physical-energy constraint then extends clearance control to prescribed endpoint manoeuvres.

5.1 Realizing the learned dynamics through two magnetic ports

Let a_j, v_j denote the mass-normalized positions and velocities of all 64 transverse finite-element modes. Their bias-field frequencies are ω_j, and their tip displacement and rotation coefficients are b_j, t_j. A transverse tip force F and a transverse field B_y give

$$\begin{aligned} \ddot{a}_j + \alpha \dot{a}_j + \omega_j^2 a_j &= b_j F + t_j U, \\ U &= E_0(\vartheta - \sin\vartheta) + \mu B_y \cos\vartheta, \\ \vartheta &= \sum_j t_j a_j. \end{aligned} \tag{44}$$

These complete modal coefficients define the plant used below; Supplement S6.1 gives its relation to the assembled finite-element matrices.

For the first two modes, write $a_i = S_i X_i$ with the diagonal scales of Section 2. Because the learned h contains momentum quartics, velocity determines Y through the implicit relation $h_Y(X, Y) = \mathrm{diag}(S_i)^{-1} v_{1:2}/\omega_0$. If $h_{YY} \succ 0$, this relation is locally invertible. Differentiating it along $\dot{X} = \omega_0 h_Y$, $\dot{Y} = -\omega_0 h_X$ gives the desired physical acceleration and hence the feedback

$$\begin{aligned} a_{i,tt}^{\mathrm{des}} &= S_i \omega_0^2 (h_{YX} h_Y - h_{YY} h_X)_i, \\ \begin{pmatrix} F \\ U \end{pmatrix} &= A_{\mathrm{mag}}^{-1} \left(a_{1:2,tt}^{\mathrm{des}} + \alpha v_{1:2} + \mathrm{diag}(\omega_1^2, \omega_2^2) a_{1:2} \right), \\ A_{\mathrm{mag}} &= \begin{pmatrix} b_1 & t_1 \\ b_2 & t_2 \end{pmatrix}. \end{aligned} \tag{45}$$

Substitution into (44) enforces the desired acceleration. Differentiating the velocity relation once more and using invertibility of h_{YY} then recovers both canonical equations.

The force and moment in (45) are physically realized by the local field $B(r) = (B_0, B_y) + G(r - r_{\mathrm{tip}})$, where

$$\begin{aligned} B_y &= \frac{U - E_0(\vartheta - \sin\vartheta)}{\mu \cos\vartheta}, \\ G &= \frac{F}{\mu} \begin{pmatrix} -\sin\vartheta & \cos\vartheta \\ \cos\vartheta & \sin\vartheta \end{pmatrix}. \end{aligned} \tag{46}$$

For the dipole $m = \mu(\cos\vartheta, \sin\vartheta)$, direct multiplication gives $Gm = (0, F)$ and $\|G\|_2 = |F|/\mu$. Symmetry and zero trace of G satisfy the local source-free curl and divergence constraints. The construction requires independent field and gradient authority, observed modal and tip states, and $|\vartheta| < \pi/2$; it does not assume independently actuated mechanical modes.

Theorem 4. Full-model realization of the certified torus. *Suppose the learned Hamiltonian has an analytic embedded two-torus with physical angular frequency $\Omega = 2\pi\omega_0\varpi$. Assume $h_{YY} \succ 0$ on a domain containing this torus with convex Y-fibres for each X, A_{mag} is invertible and $\alpha > 0$. If the lifted motion stays in the domain $|\vartheta| < \pi/2$ of the available commands, feedback (45)–(46) gives an embedded invariant two-torus of the complete finite-dimensional plant. For each low-mode orbit on that torus, higher-mode deviations from its unique quasiperiodic response decay exponentially along closed-loop trajectories that remain in the stated command domain.*

Proof. The feedback reproduces the low-mode Hamiltonian exactly. Its demands F, U therefore depend only on the torus phase. With Fourier convention $e^{ik\cdot\phi}$, each higher-mode response is

$$a_{j,k}^{*} = \frac{b_j F_k + t_j U_k}{\omega_j^2 - (k\cdot\Omega)^2 + i\alpha(k\cdot\Omega)}, \qquad j \geq 3. \tag{47}$$

The denominator is nonzero and its inverse is bounded on real frequencies: at zero its real part is positive, and away from zero its imaginary part is nonzero. Analytic forcing thus gives an analytic response on a possibly smaller strip. Positive definiteness on each convex Y-fibre makes $Y \mapsto h_Y(X, Y)$ strictly monotone and injective. The physical low-mode coordinates therefore preserve the torus embedding. Differences between high-mode solutions satisfy homogeneous damped oscillator equations and decay exponentially.

This lift is derived here from the model and the torus supplied by the external KAM theorem. Its physical angular frequency is approximately $(11.41719351, 70.24034116)$ rad/s. The positive-measure family remains a statement about the

Table 7. Rigorous bounds for the COMSOL-integral learned model. Startup means zero higher-mode positions and velocities, with the low state on the certified torus.

Quantity	Upper bound
Centreline on response graph	1.922696 mm
Centreline including startup	1.970263 mm
Transverse field including startup	0.921665 mT
Gradient norm including startup	0.02572538 T/m

four-dimensional low subsystem. Without sustained input, the physical beam instead converges to its globally asymptotically stable equilibrium (Supplementary Proposition S2).

For the COMSOL integral fit, interval bounds give $h_{YY} \succeq 0.9999999983I$ on the real 0.3 box and exclude zero from $\det A_{\mathrm{mag}}$. The response and startup estimates in Supplement S6.2 include all 62 higher modes and the complete nonlinear forcing remainder. Cubic-Hermite spatial bounds yield the whole-centreline and command limits in Table 7. With the low state on the exact certified torus and all higher positions and velocities initially zero, the slope remains below 0.155261 rad. Hence $\cos\vartheta > 0.987971$, closing the feedback-domain condition for every phase and all time.

Figure 2 shows the full-state realization. Over 12 s, the low-mode state agrees with an independent Hamiltonian integration to within 1.54×10^{-10} in canonical coordinates. Online COMSOL feedback provides a second implementation check: refining its fixed step from 1 to 0.125 ms reduces the four-second tip RMSE against the independent full-state solver from 147.852 to 3.848 μm. The analytic and interval estimates supply the all-time bounds in Table 7; Supplement S6.1 reports the complete refinement study.

5.2 Frequency continuation and the clearance boundary

To test where driven motion changes character, we vary a single sinusoidal transverse field from 1 to 20 mT and its frequency ratio $r = \omega_f/\omega_0$ from 0.7 to 1.3. The two-mode physical model is evaluated at 1,029 independently initialized points and 686 upward/downward continuation points. Each run discards 128 forcing periods and diagnoses the next 64 periods. All tested responses converge to forcing-periodic motion; their numerical Floquet spectral radii range from 0.803809 to 0.889055. These observations locate no quasiperiodic or chaotic transition within the sampled protocol. Large-deflection extrapolations are separately flagged, and this finite scan does not establish global absence of chaos or breakup of the conservative KAM tori.

A distinct practical boundary appears when periodic bending reaches the 4 mm clearance. Solving for periodic responses with all 64 modes gives threshold amplitudes of approximately 7.5005, 1.91685 and 8.7909 mT at $r = 0.9$, 1 and 1.1, respectively. These are numerical steady-response thresholds; the specified startup ramp can cross the clearance earlier. Supplement S5.9 reports the initial states, continuation protocol, transient thresholds, modal and temporal refinement, and independent full-model checks. The resonance boundary motivates a work constraint that covers transients as well as settled motion.

5.3 Clearance beyond the invariant torus

Tracking a prescribed path need not keep the state on a conservative torus. We therefore derive a clearance condition from the physical energy itself. For the clamped small-slope beam, $w(s) = \int_0^s (s-\xi) w_{ss}(\xi)\, d\xi$. Since the static magnetic energy is nonnegative, Cauchy–Schwarz gives

$$|w(s,t)|^2 \le \frac{2s^3 E(t)}{3EI}, \qquad E_c = \frac{3EIc^2}{2L^3}, \tag{48}$$

where c is the corridor half-width less the rod radius. Thus $E < E_c$ guarantees clearance of the entire centreline and its radius tube. This argument applies to sufficiently regular continuum solutions satisfying the power identity and separately to conforming Hermite finite-element solutions.

For a single transverse magnetic input, its actual power is $a_p B_y$, with $a_p = \mu \cos\vartheta\, \dot\vartheta$. Starting from a valid bound $\overline{E}_0 \ge E(0)$, let $\dot W = a_p B_y$ and $W(0) = \overline{E}_0$. Nonnegative dissipation gives $E \le W$. Choose $\overline{E}_0 < E_{\max} < E_c$, and project the requested field onto

$$|B_y| \le B_{\max}, \qquad a_p B_y \le \kappa(E_{\max} - W), \quad \kappa > 0. \tag{49}$$

Zero field is feasible while $W \le E_{\max}$. Moreover, (49) implies $E_{\max} - W(t) \ge (E_{\max} - \overline{E}_0)e^{-\kappa t}$. Together with (48), this proves all-time clearance under exact port measurement and command application. The result applies to driven trajectories through their supplied work, including motions away from a KAM torus; additional work ports or measurement errors must enter the same budget.

For a 4 mm net clearance, $E_c = 23.8565$ nJ. An energy budget $E_{\max} = 0.95E_c$ bounds the centreline by 3.898718 mm in continuous time. Under a 10 mT sinusoidal request, the sampled peak falls from 5.14478 mm without the governor to 3.14362 mm with it; online COMSOL gives 3.143643 mm. Supplement S7.1 supplies the power derivation, an open-loop field-variation bound and refinement data.

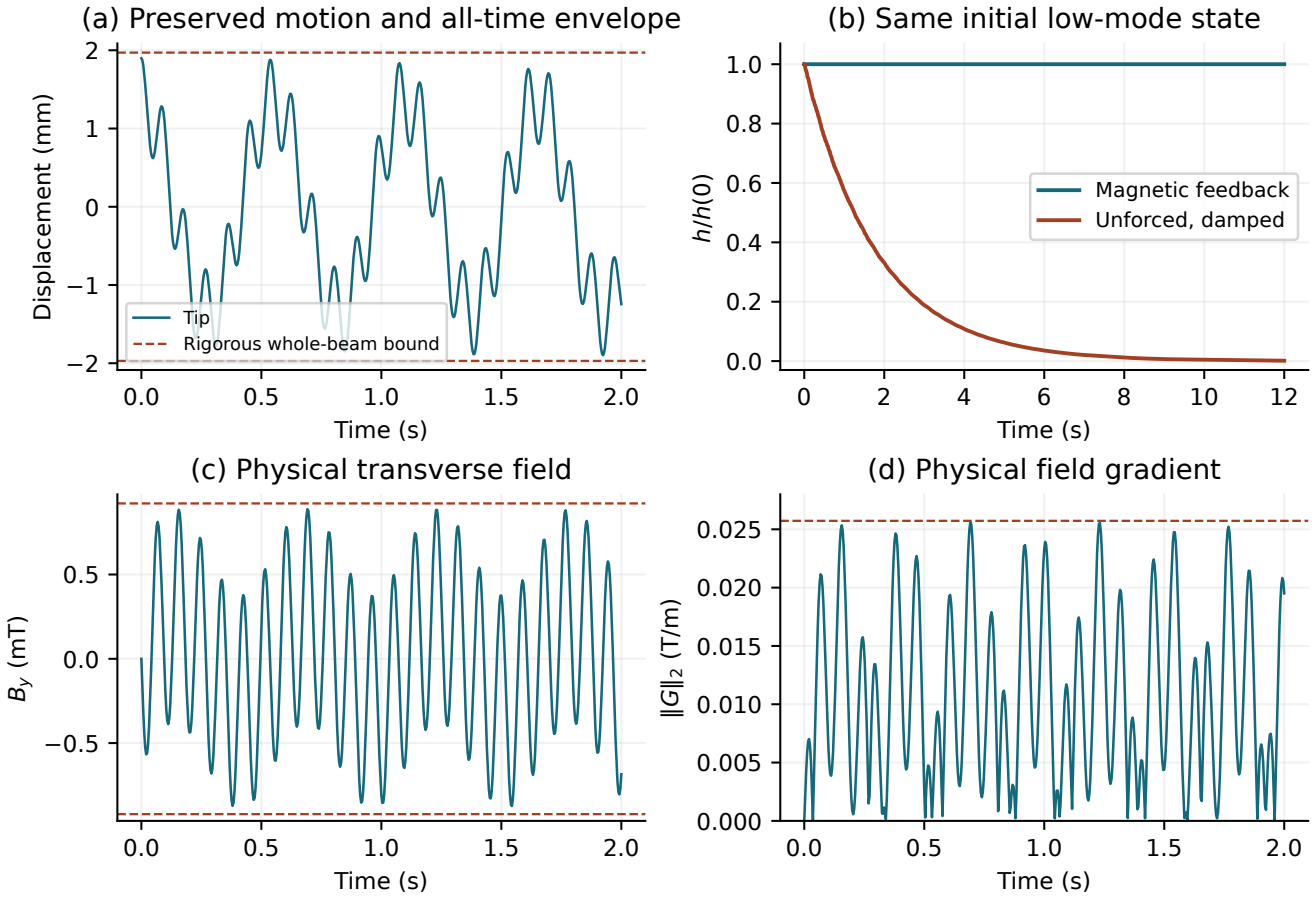


Figure 2. Realizing the certified learned dynamics on all 64 modes. The higher modes start at zero. Dashed envelopes include the startup transient and bound every phase and centreline point. Magnetic field and gradient jointly supply the force and torque; the passive comparison uses the same initial state without this feedback. Panels (a,c,d) show 2 s and panel (b) shows 12 s.

5.4 Nonzero sensing, initialization and actuation tolerances

The extremely narrow action tube in the KAM measure estimate is a proof domain, rather than an initialization tolerance. A separate physical-energy argument yields a nonzero operating tolerance about the exported finite Fourier reference. In the frozen full nodal model, let $Q(e) = \frac{1}{2}e_v^{\mathsf{T}} M e_v + \frac{1}{2}e_q^{\mathsf{T}} K_b e_q$. The initialization set $Q(e) \le 5$ pJ implies a tip-position error below $56.006\,\mu$m; this is a projection of an energy ellipsoid, not a box of independently admissible coordinate errors.

For the two work ports, bound tip-angle, angular-rate and tip-velocity measurement errors by 1 mrad, 0.01 rad/s and 0.1 mm/s. Allow 1% multiplicative error in both applied inputs, together with additive errors of 0.5 nN in force and $0.05\,\mu$T in field. A worst-case measured-power bound and a dissipation-based reserve of 0.096780 nJ cover these additive errors over 4 s. The resulting 22.66 nJ energy cap bounds the whole centreline below 3.898405 mm throughout that interval. Supplement S7.5 gives the proof, directed-rounding certificate, actuator derating and full-model refinement. The governor can alter the requested motion substantially: this guarantee concerns clearance under the stated errors and does not preserve exact torus membership or certify tracking accuracy.

5.5 Endpoint tracking and practical scope

We next combine axial base translation with magnetic bending to trace an S-shaped endpoint path (Figure 3). The tip advances 48 mm over a 12 s stroke with lateral extrema ± 1.5 mm. Axial acceleration ends before bending begins, and the model has no axial drag or guide contact. With base kinetic energy and work accounted for separately, the transverse governor bounds clearance in the straight corridor (Supplement S7.6). The prescribed S shape is the moving endpoint trajectory.

The controller uses the two-mode Hamiltonian fitted to full-FE COMSOL trajectories, its full gradient and Hessian, observed low-mode projections and measured tip feedback, with coefficients fixed throughout the task. Table 8 compares controllers on the same trajectory and plant with a 20 mT field limit. The learned controller achieves $6.419\,\mu$m RMS error, while the known-physics inverse performs better on

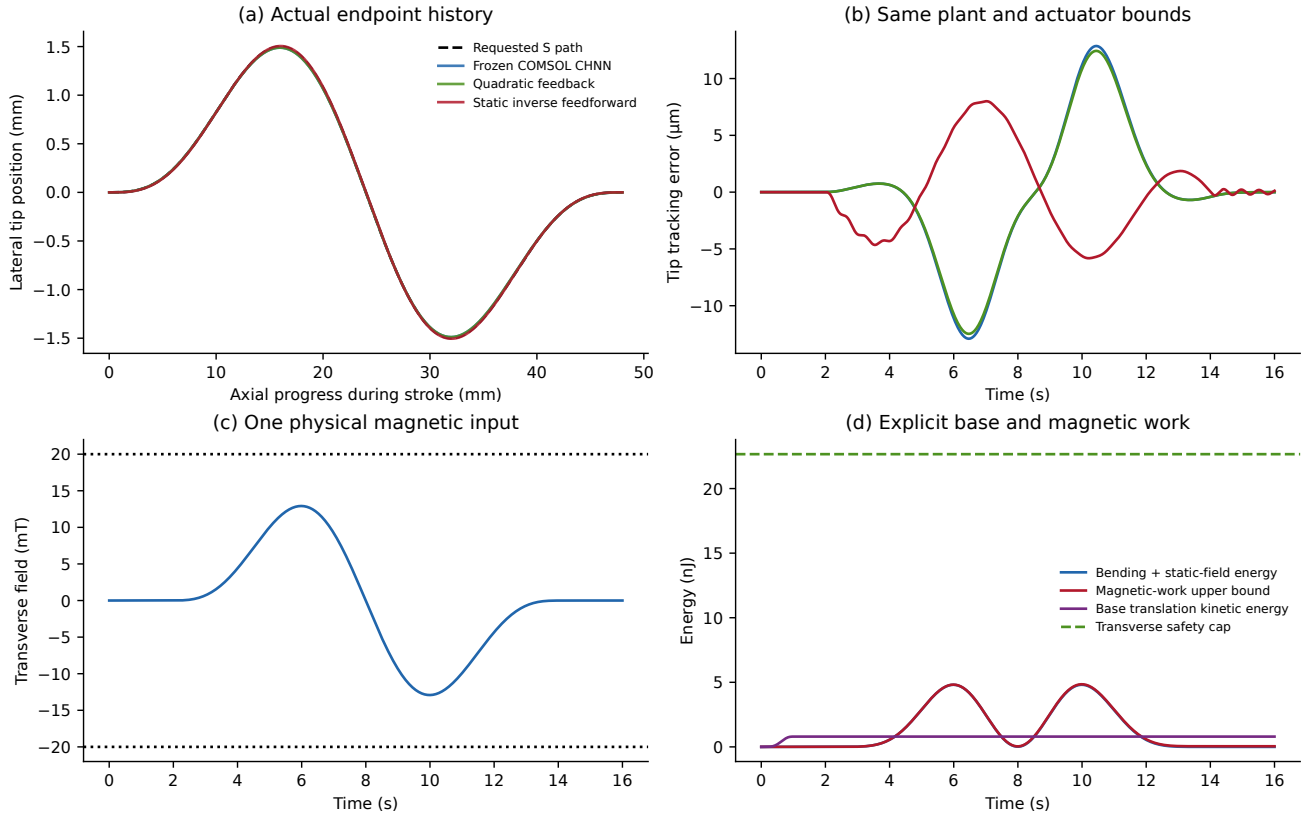


Figure 3. S-shaped endpoint tracking through axial base feed and transverse magnetic actuation. The same full beam and input limits are used for the matched controls. Base work is included in the energy balance.

Table 8. Matched S-path controls. RMS errors use the 12 s stroke; maxima use the complete 16 s simulation.

Control	RMS (μm)	Max. (μm)
CHNN, tip feedback	6.419	12.907
Quadratic, tip feedback	6.207	12.476
Static inverse feedforward	4.124	8.005
CHNN, modal-output feedback	25.308	41.544

this slow nominal path. The modal-output comparison shows the benefit of using the measured tip in feedback.

The learned command peaks at 12.9177 mT and the centreline at 1.48892 mm. The maximum recorded work-based energy estimate is 4.84997 nJ, and the governor remains inactive for this request. An independent COMSOL replay of the field agrees with the predicted tip to within 4.038×10^{-10} m. The replay tests this task's mechanical response; the torus experiment above additionally tests online feedback.

The resulting design guidance is to preserve fitted energy derivatives, check actuation rank and momentum invertibility, and budget all input work. The envelopes require the stated state estimates and local field/gradient authority. Device implementation requires calibrated hardware and verified error bounds; partial sensing, large rotations, flow and wall contact need corresponding extensions.

Safety for the jointly fitted variable-inertia model. The planar joint fit has its own power and safety argument (Supplementary Section S3.17). Its single magnetic port is $g(q) = d\cos(d^{\mathsf{T}}q)$, with measured output $y = g^{\mathsf{T}}q'$. The fitted real energy and nonnegative Rayleigh head satisfy $H'_\theta = -\widehat{\gamma}P^{\mathsf{T}}G_Q^{-1}P + uy$. The certified initial potential error supplies an upper bound for the true energy. A scalar input-work filter then guarantees clearance under the true plant's nonnegative dissipation and exact port sensing and actuation. A separate energy barrier keeps the trajectory within the network's configuration box, and a bending comparison accounts for stored-buffer rounding in the finite-angle centreline bound. This proof uses the same chart and metric as the new fit. Rank-one actuation does not supply the arbitrary two-mode compensation used in the earlier two-port realization; the new theorem establishes safety without that controllability claim.

6 Conclusion

We connected an identified magnetic energy to numerical prediction, nonlinear persistence and physical actuation in four steps. The complex representation and classical central symplectic update preserve the model's structure; the finite-feature approximation analysis carries nonzero identification error into the KAM hypotheses; and magnetic feedback realizes the resulting finite-amplitude torus in the damped full beam. The model-specific realization and lift yield explicit centreline, field and gradient bounds. Input-work control extends clearance guarantees to driven motions beyond that torus, with explicit nonzero initialization, sensing and actuation tolerances over a prescribed finite horizon. The frequency study identifies a resonance-induced clearance boundary; no chaotic transition is detected under its sampled protocol.

The finite-feature numerical comparisons define the scope of that result: real and complex implementations have equivalent resolved accuracy, and the known-physics inverse performs best on the slow nominal tracking task. Extending the analysis to partial sensing, calibrated actuators and three-dimensional contact requires corresponding mechanical models and error bounds.

The fixed-mass end-to-end extension learns two complex fully connected hidden layers and a position-dependent PSD damping head. Pole constraints retain a declared analytic domain. Weak momentum windows and physical projection bounds specify the errors needed to infer conservative accuracy. Direct interval verification bounds the frozen function locally and supplies counterexamples on larger boxes. Complete normal forms establish local nondegeneracy, and separate a posteriori certificates give learned and nominal finite-amplitude tori at the same frequency. Their phase-paired distance is bounded, but their amplitudes differ; this is not a same-initial-state prediction theorem. The earlier full-beam realization retains its own fitted energy and two-port assumptions.

For variable inertia, a planar inextensible Kirchhoff restriction retains rotary inertia, configuration transport and the kinetic contributions to the normal form. Its actual quadrature conventions retain the verified four-jet. Native COMSOL position–velocity windows support a separate joint fit of a structured complex potential and scalar mass-proportional drag. A profiled full-batch Armijo theorem matches that objective; independent ball arithmetic certifies the frozen potential and conservative-field errors. The same decoder has an exact power identity and, for prescribed input, a time-dependent Birkhoff form and conformal second-order scheme. A work filter, initialized by the certified potential bound, proves clearance and invariance of the configuration proof box for the true reduced plant under exact sensing and actuation. Numerical closed-loop checks test safety and report the substantial remaining tracking error. This branch establishes same-chart fitting, function-error bounds, a same-checkpoint finite-amplitude conservative KAM certificate and conditional continuous-time safety. The certificate includes the full variable-inertia kinetic energy, continuous mean torsion and the infinite Fourier residual tail. A separate complex factor network now learns the dissipative term while this conservative energy stays frozen. The PSD decoder and correct Rayleigh mass factor give exact nonnegative loss and port power. A matched regularized weak-loss descent theorem and whole-box field-error certificate apply to the trained head. Four held-out trajectories and the unchanged single-port task check its numerical implementation, while retaining the saturation and tracking limits. A separate finite-horizon sampled-work theorem and uniform same-plant port bounds now admit nonzero sensing and execution errors. Its explicit reserve, true-damping and clock hypotheses remain conditions on an implementation. An enlarged continuous design and explicit quadratic remainder also bound the normalized dissipative matrix on the full configuration box; its joint coefficient ball retains a zero nonconstant block. Two-sided damping bounds inferred from the frozen head and its certified function error also supply a positive finite-horizon sampled-port safety budget for the same reference class. These constant-damping data do not demonstrate a nontrivial state-dependent law or a complex-network advantage. Validated native reconstructions now give continuous finite-Gauss IVP and learned weak-equation enclosures for the final output. A verified continuous design and explicit higher-order tails additionally turn those residuals into a conservative-gradient error bound for that same output. For the same declared conservative pair, uniform derivatives and average-torsion bounds, combined with the verified explicit-response residual, establish same-frequency invariant tori at every real point of the learned–reference homotopy. Statistical generalization, the native solver's hidden internal trajectory and correspondence to an unconstrained three-dimensional beam remain open.

Data and code availability

The supplied materials comprise two identified archives. Archive S1 is the complete computational record

dated 2 October 2026, containing native COMSOL models, construction and learning code, trajectories, fitted coefficients and interval records. Archive S2, `S2_citation_revision_20261008.zip`, contains the revised article and Supplement, the additional stability and tolerance studies, frozen inputs, raw numerical outputs and reproduction commands. The deep-learning experiments additionally supply its frozen networks, training and held-out results, code, numerical domain and derivative diagnostics, the native COMSOL joint-fit results, and the separately bound conservative KAM inputs. The accompanying `reproducibility/ARTIFACT_INDEX.md` identifies the archives and maps results to files and checksums; its source map separates full-FE identification from the two-mode integral data used for the KAM proof.

Declarations

This work reports mathematical analysis and computer-generated mechanical data. No human, animal or in-orbit experiment was performed.

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